

Traverse Computations

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Field Exercise

Aim:

Determine the distanced and internal horizontal angles between Traverse Points.

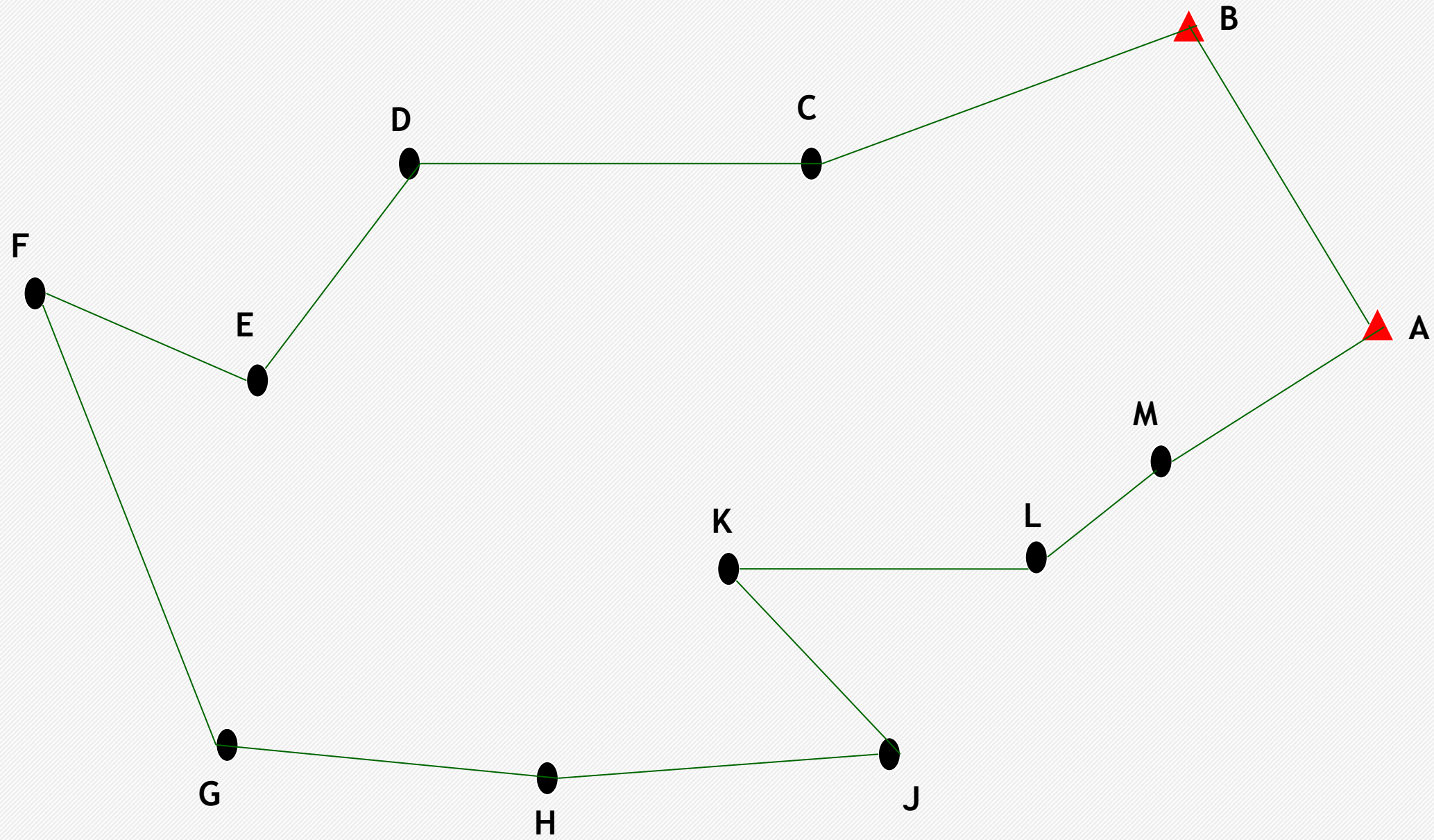
Equipment:

Total Station

Given Data:

Coordinates of Points A and B.

NICE Traverse



Traverse

A Traverse consists of a connected series of lines on Earth's surface, the length and bearings of which have been determined.

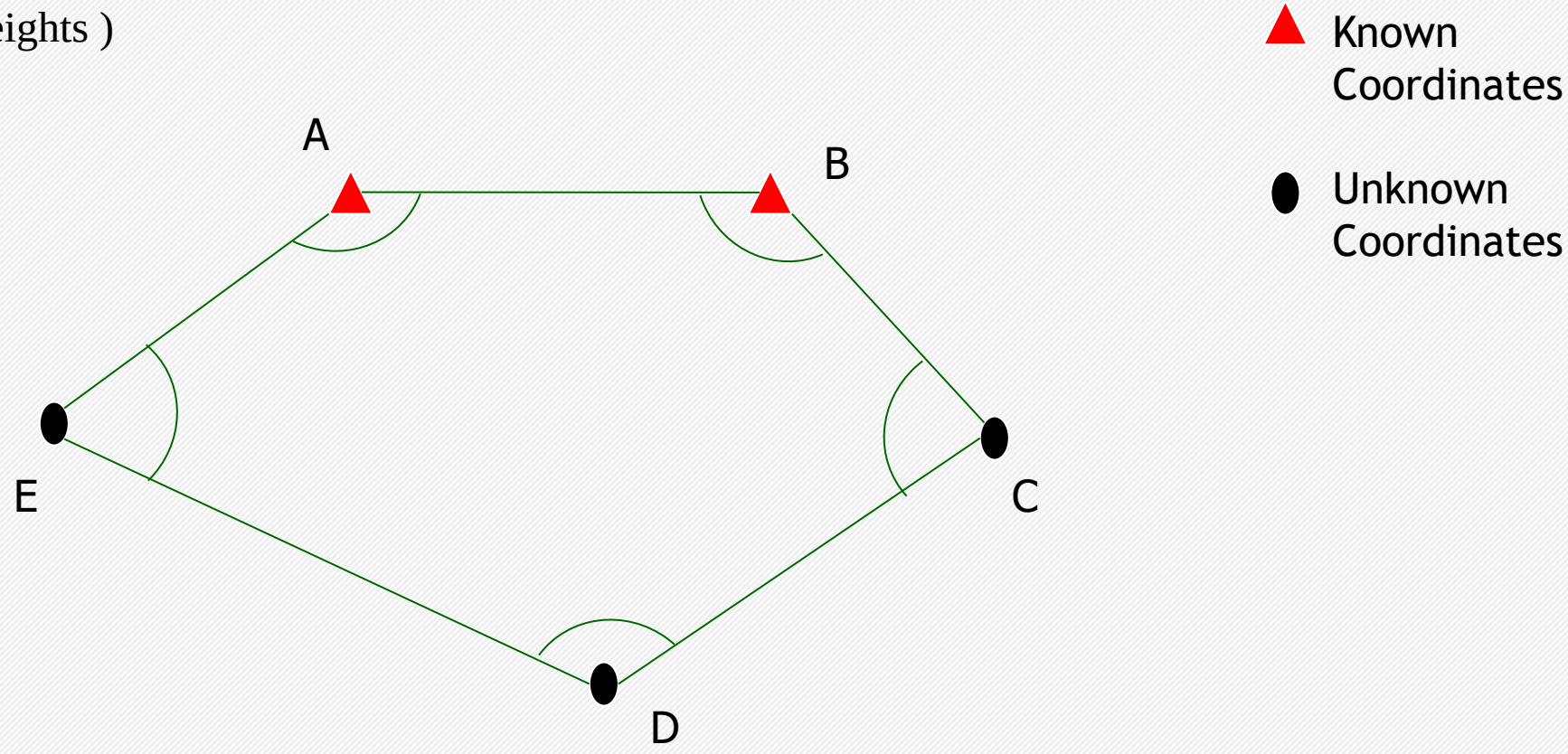
Following measurements are made:-

a. Horizontal Angles

b. Vertical Angles (For Heights)

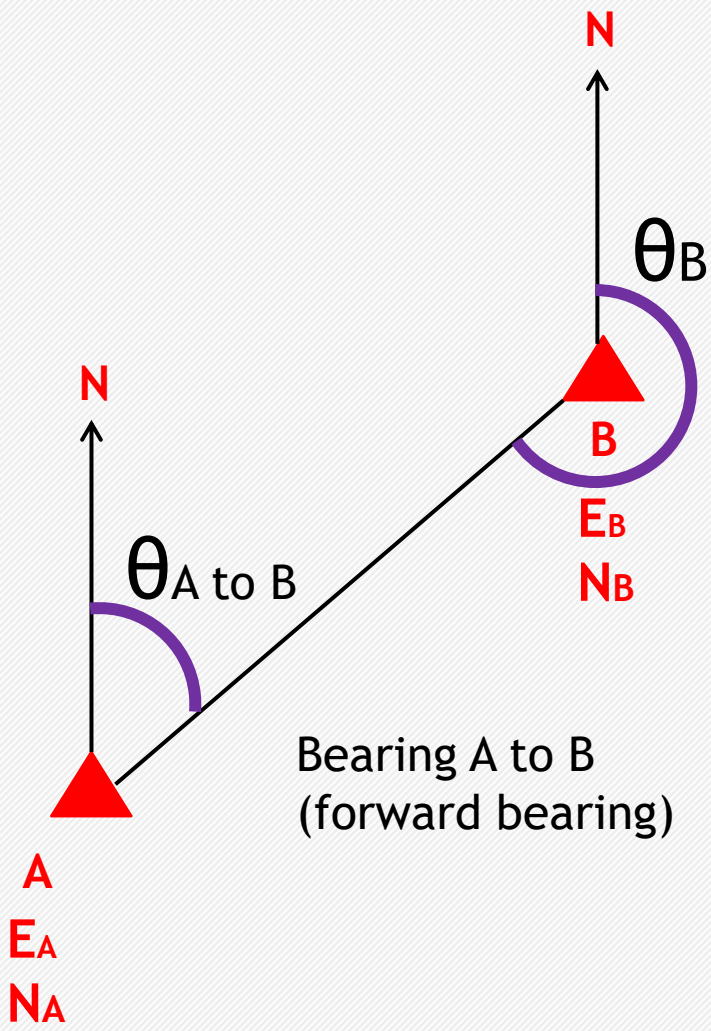
c. Distances

d. Bearings



Computations

Bearings



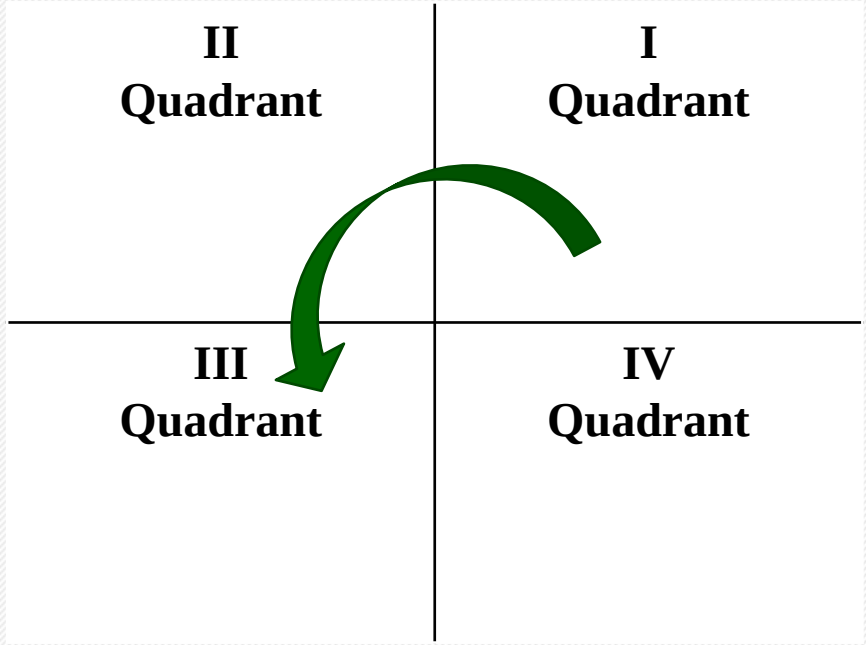
Forward Bearing - Backward Bearing = 180º

If:
Forward Bearing > 180º
Backward Bearing = Forward Bearing - 180º

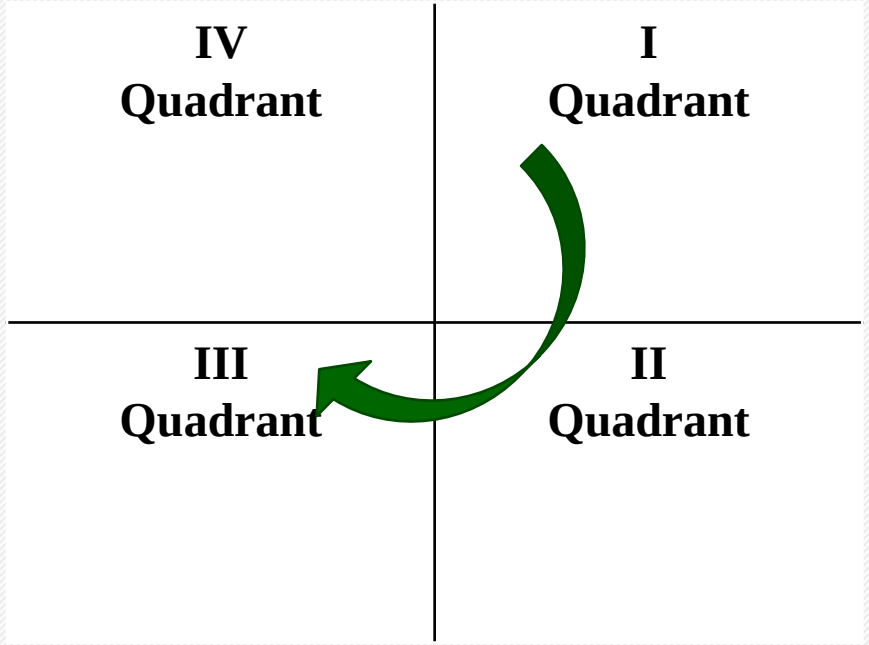
If:
Forward Bearing < 180º
Backward Bearing = Forward Bearing + 180º

Computations

Quadrant Systems



Conventional



Surveying

Computations

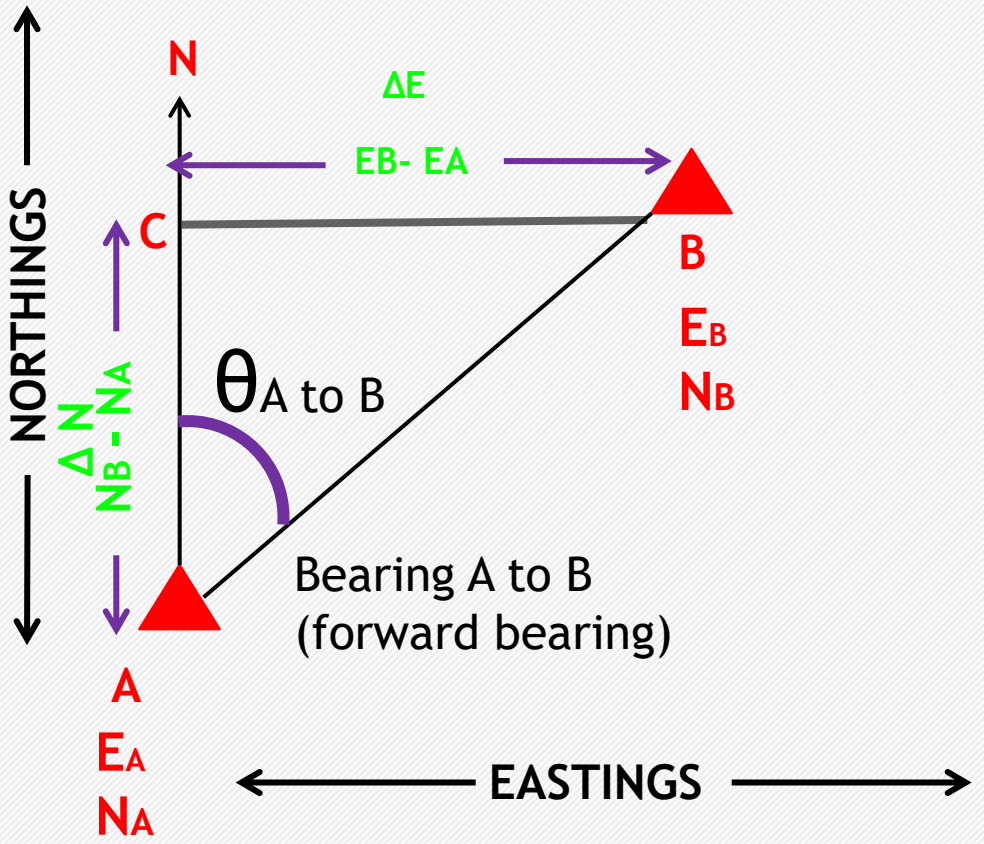
Bearing Computation

Opening Bearing

Always calculated from known points.

Forward Bearing

Bearing A to B = $\tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ B - Easting\ of\ A}{Northing\ of\ B - Northing\ of\ A}$



Computations

Bearing Computation

Opening Bearing

Always calculated from known points.

Forward Bearing

$\theta = \tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ B - Easting\ of\ A}{Northing\ of\ B - Northing\ of\ A}$

1st Quadrant

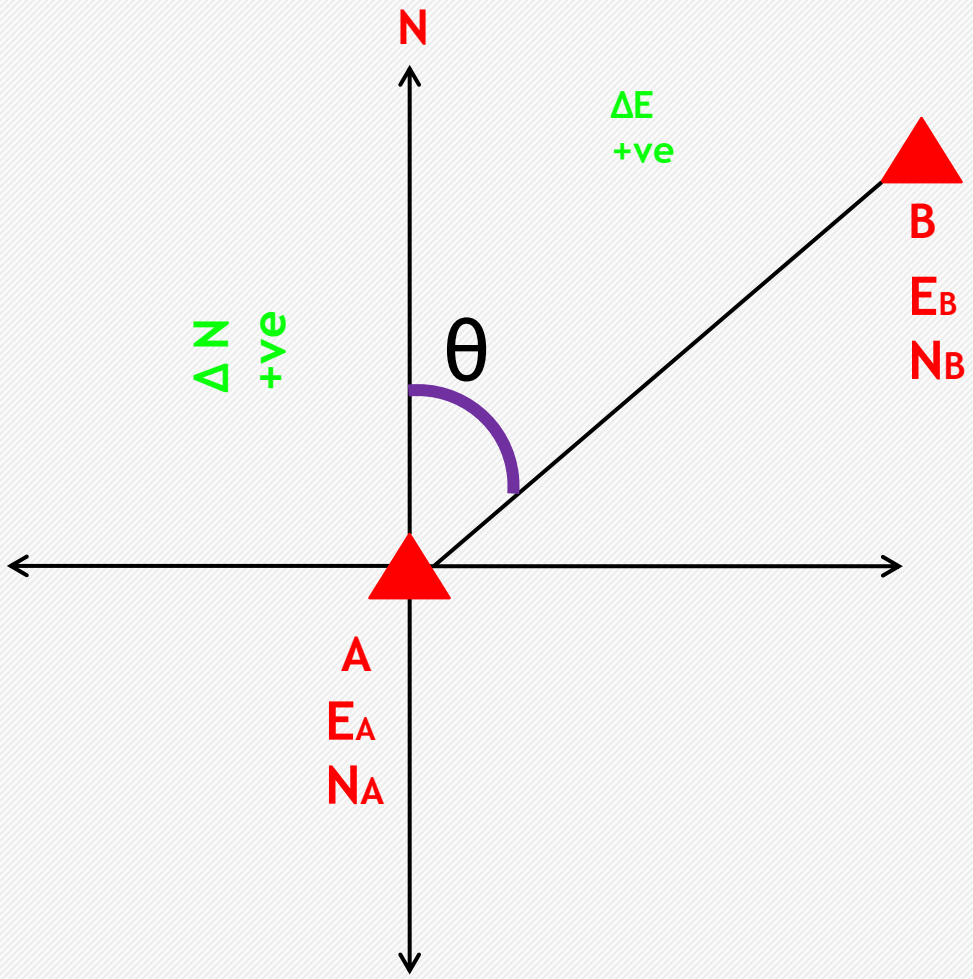
$E_B - E_A = +\Delta E$

$N_B - N_A = +\Delta N$

$\theta = Tan^{-I}\frac{+\Delta E}{+\Delta N}$

$\theta = +iv$

Bearing A → B = θ



Computations

Bearing Computation

Opening Bearing

Always calculated from known points.

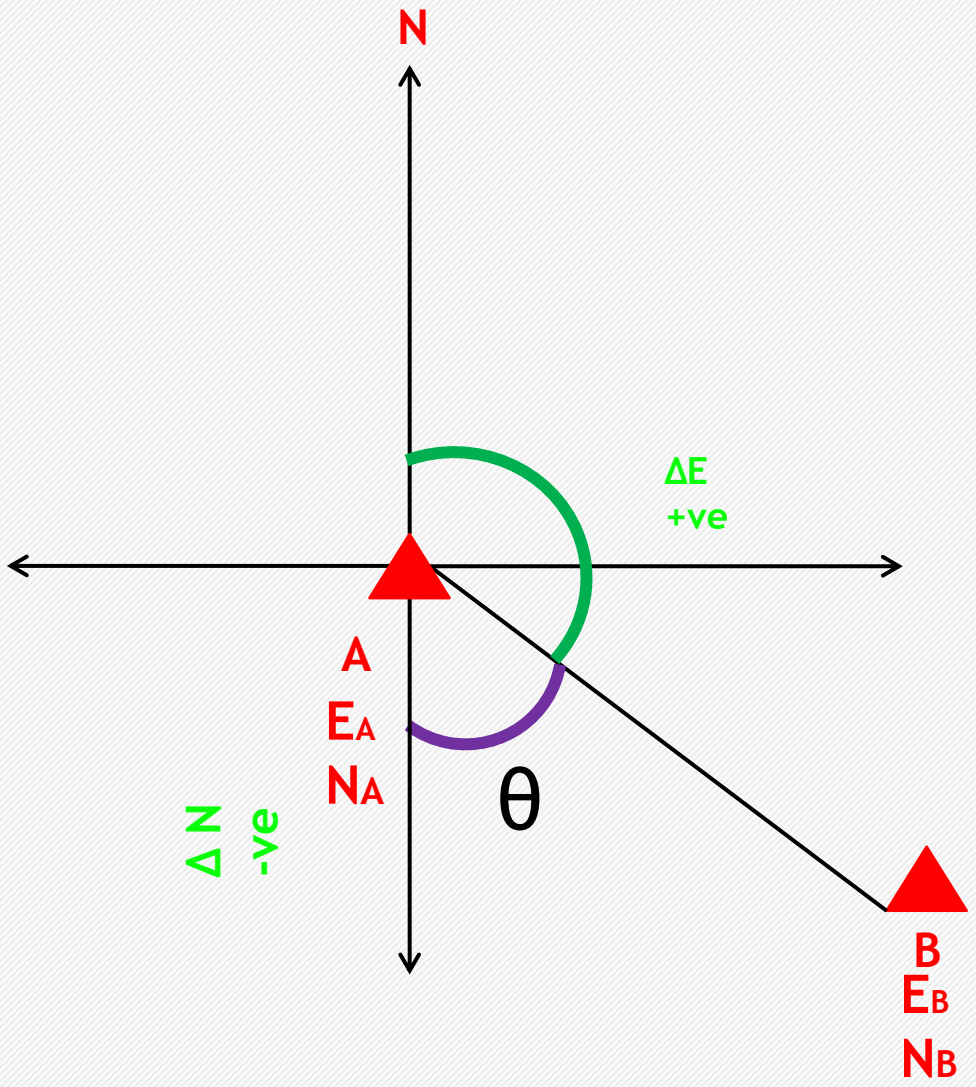
Forward Bearing

$$\theta = \tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ B - Easting\ of\ A}{Northing\ of\ B - Northing\ of\ A}$$

2nd Quadrant

$$E_B - E_A = +\Delta E$$
$$N_B - N_A = -\Delta N$$
$$\theta = Tan^{-I}\frac{+\Delta E}{-\Delta N}$$
$$\theta = -iv$$

Bearing A → B = 180-θ



Computations

Bearing Computation

Opening Bearing

Always calculated from known points.

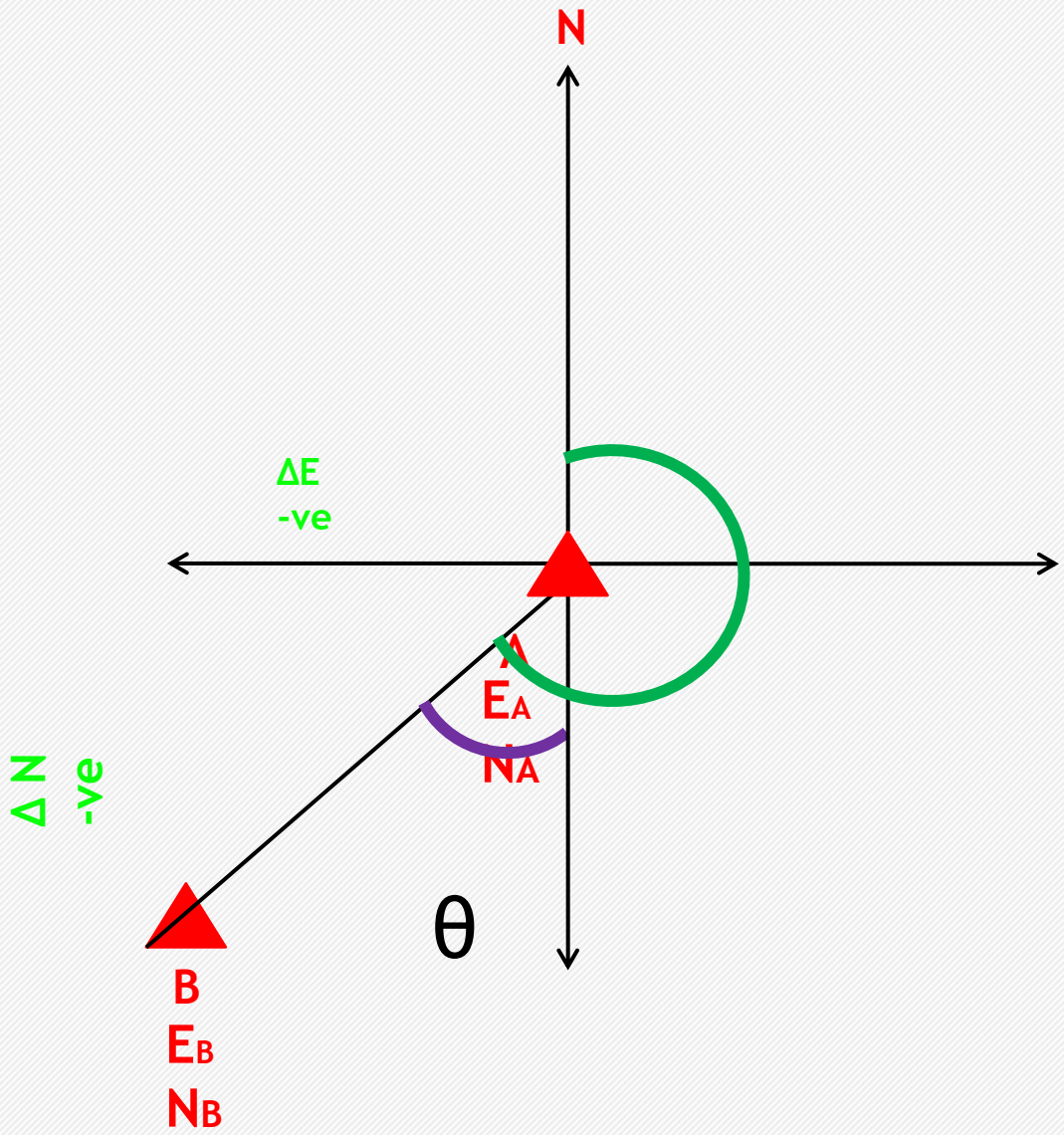
Forward Bearing

$$\theta = \tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ B - Easting\ of\ A}{Northing\ of\ B - Northing\ of\ A}$$

3rd Quadrant

$$E_B - E_A = -\Delta E$$
$$N_B - N_A = -\Delta N$$
$$\theta = Tan^{-I}\frac{-\Delta E}{-\Delta N}$$
$$\theta = +iv$$

Bearing A → B = 180+θ



Computations

Bearing Computation

Opening Bearing

Always calculated from known points.

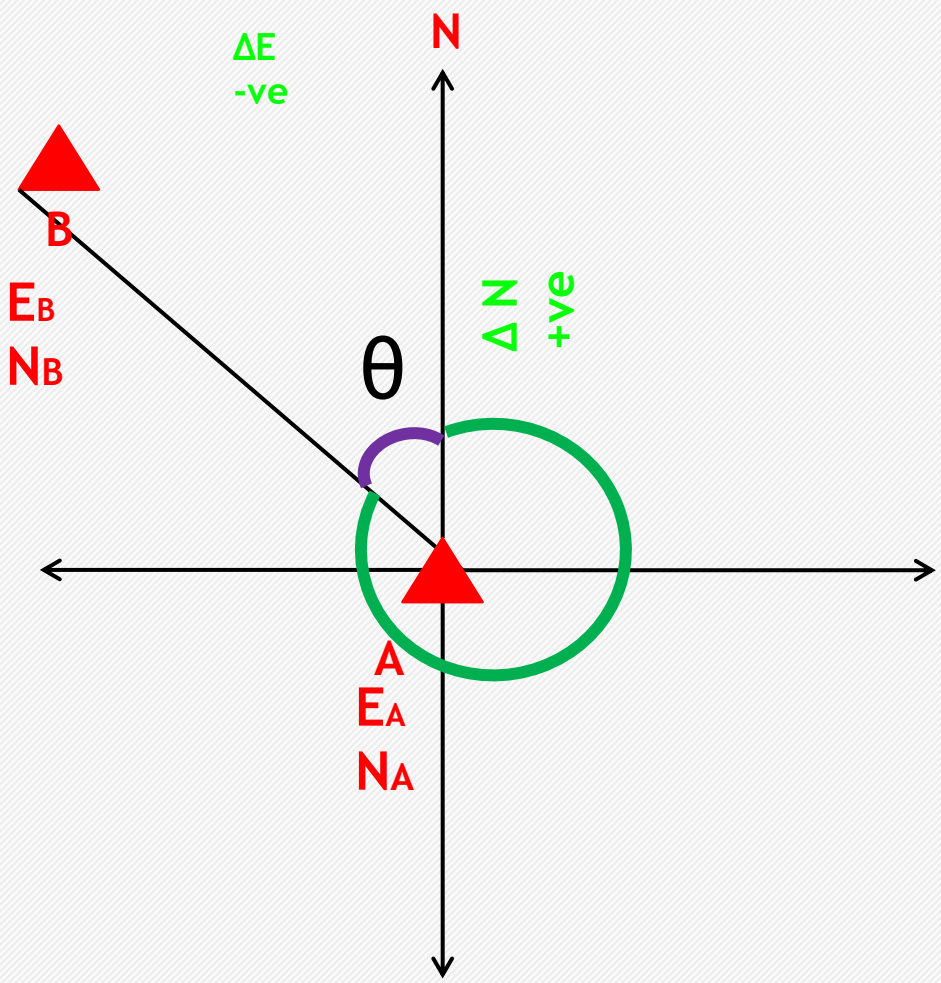
Forward Bearing

$$\theta = \tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ B - Easting\ of\ A}{Northing\ of\ B - Northing\ of\ A}$$

4th Quadrant

$$E_B - E_A = -\Delta E$$
$$N_B - N_A = +\Delta N$$
$$\theta = Tan^{-1}\frac{-\Delta E}{+\Delta N}$$
$$\theta = -iv$$

Bearing A → B = 360-θ



Computations

Bearing Computation

Opening Bearing

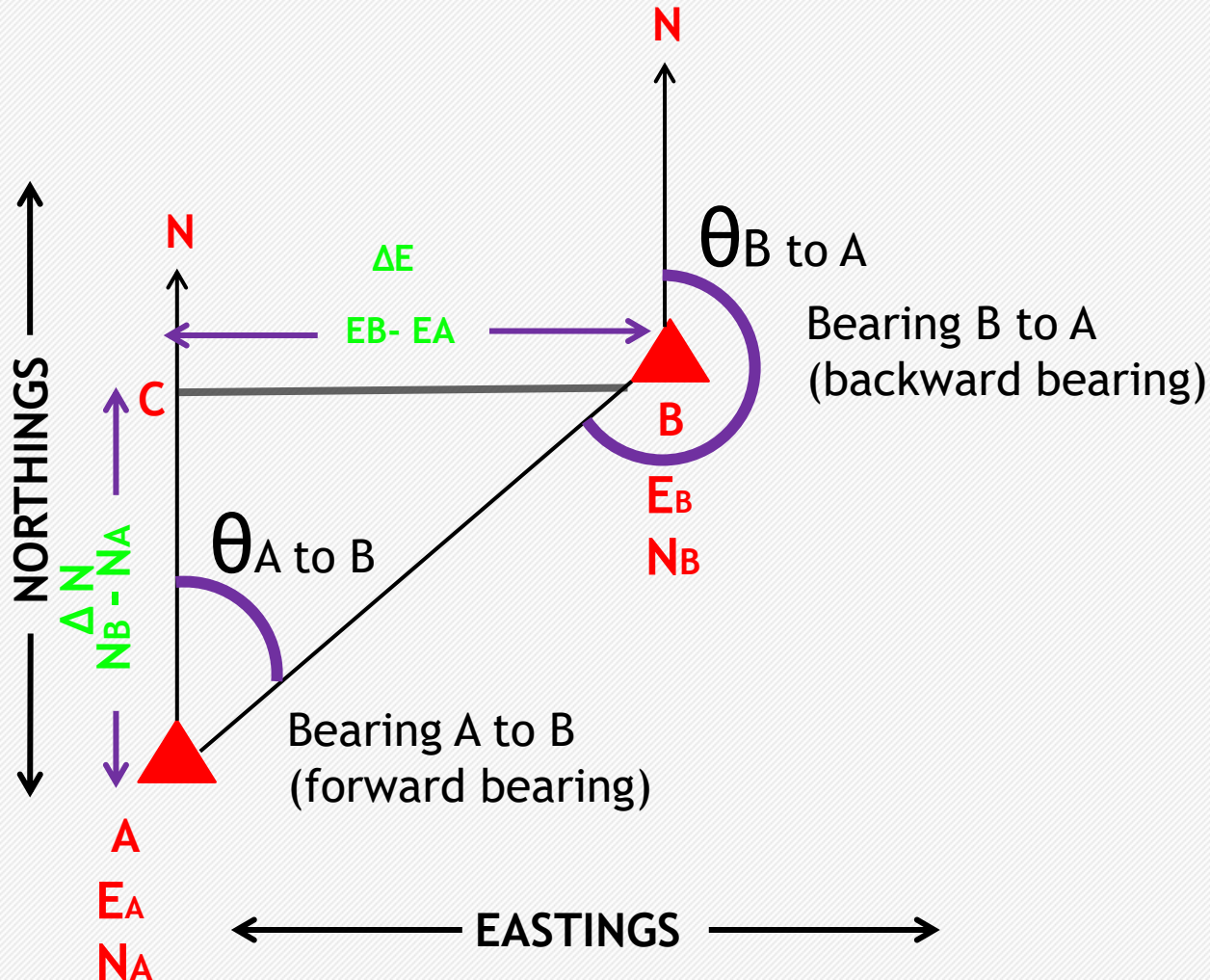
Always calculated from known points.

Forward Bearing

Bearing A to B = $\tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ B - Easting\ of\ A}{Northing\ of\ B - Northing\ of\ A}$

Backward Bearing

Bearing B to A = Bearing A to B $\pm\ 180^0$



Computations

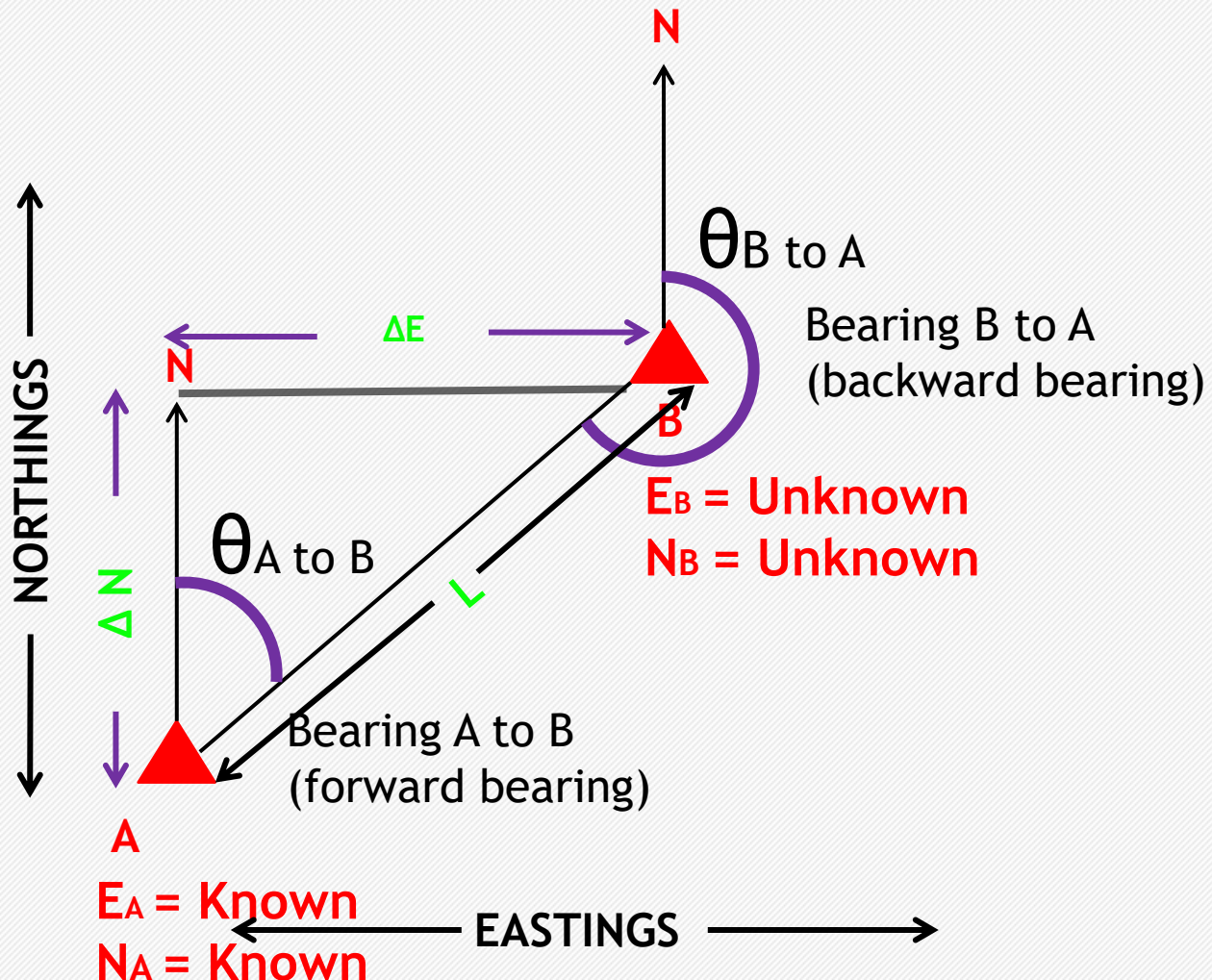
Co-ordinates

$\Delta E = L \sin\theta_{A \text{ to } B}$

$\Delta N = L \cos\theta_{A \text{ to } B}$

$E_B = E_A + \Delta E$

$N_B = N_A + \Delta N$



LET'S PRACTICE



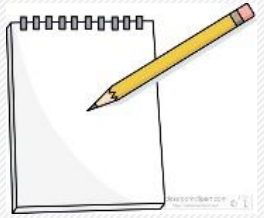
Computations



PLEASE TAKE NOTES

Check List

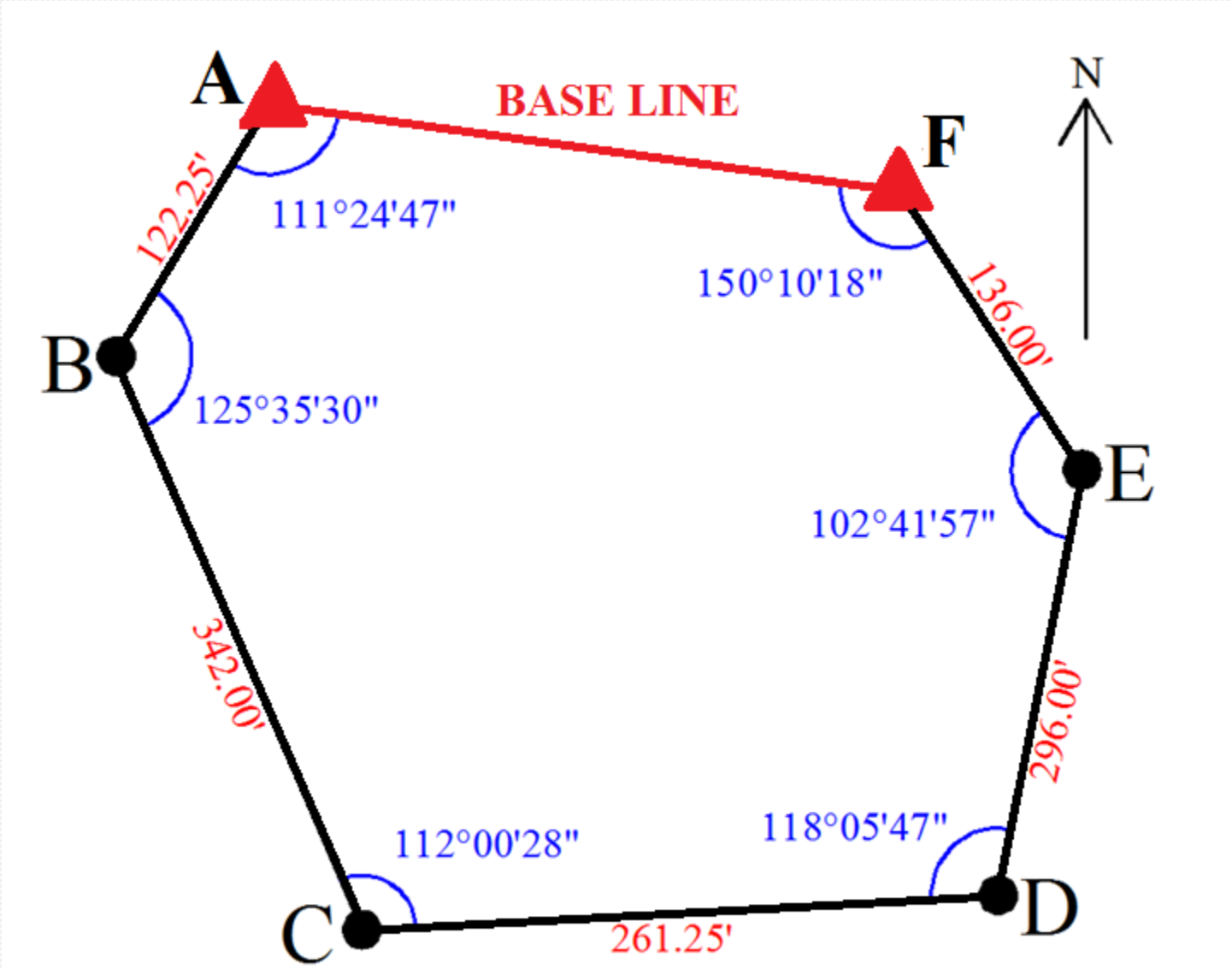






?

Computations



Coordinate of point A

E_A=4500ft
N_A=5300ft

Coordinate of point F

E_F =4860.63ft
N_F=5236.41ft

Computations

Opening Bearing

Forward Bearing

Bearing A to F = $\tan^{-1}\frac{\Delta E}{\Delta N} = \frac{Easting\ of\ F - Easting\ of\ A}{Northing\ of\ F - Northing\ of\ A}$

Bearing A to F = $\tan^{-1}\frac{4860.3-4500}{5236.41-5300}$

Bearing A to F = $\tan^{-1}\frac{4860.63-4500}{5236.41-5300}$

Bearing A to F = $\tan^{-1}\frac{360.63}{-63.59}$

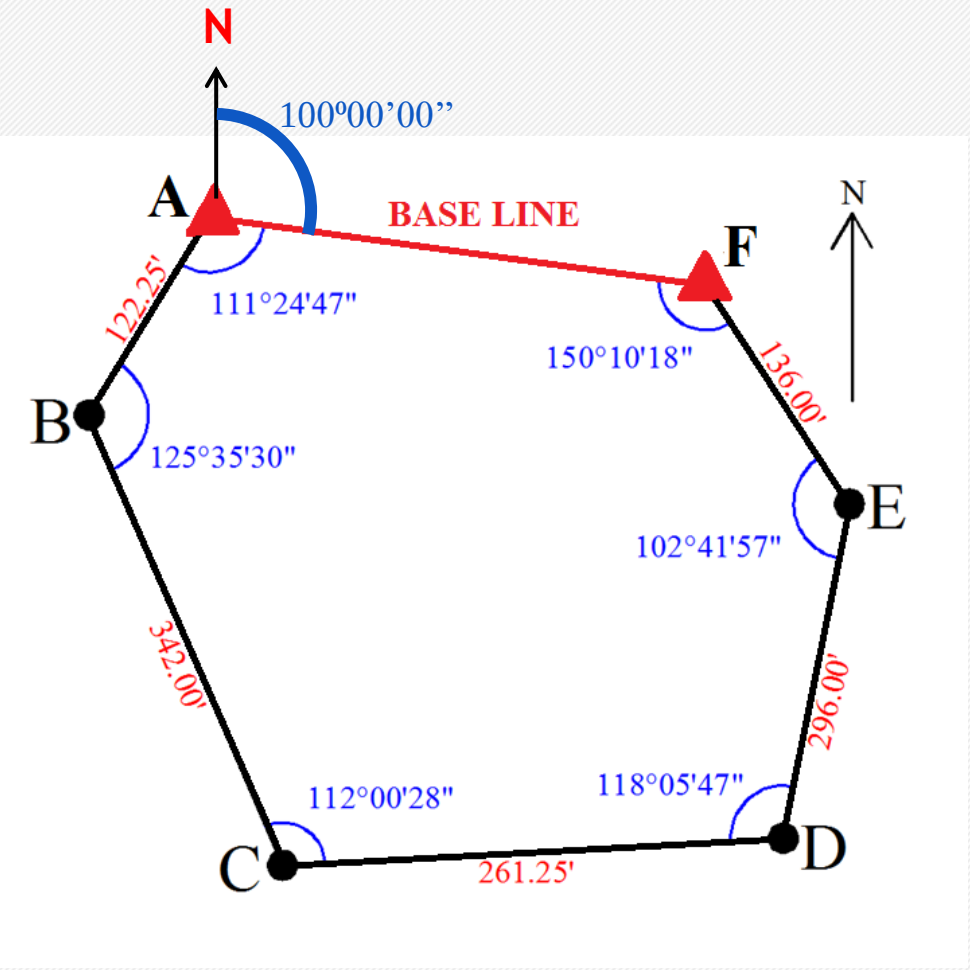
Bearing A to F = -79.99981495º

Bearing A to F = -79º59’59.33”

Bearing A to F ≈ -80º00’00”

Bearing A to F = -80º00’00” + 180º00’00”

Bearing A to F = 100º00’00”



Computations

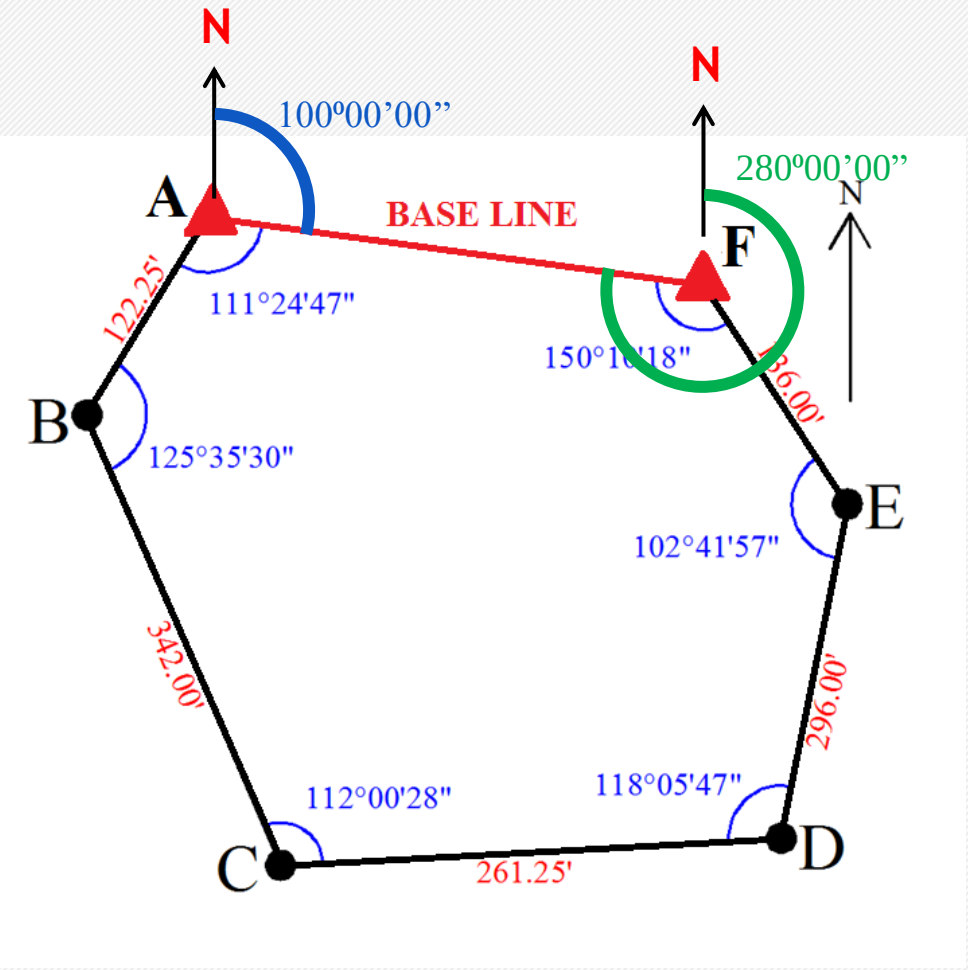
Backward Bearing

Bearing F to A = ?

Bearing F to A = Bearing A to F $\pm$ 180^0

Bearing F to A = 100º00’00” + 180^0

Bearing F to A = 280º00’00”



Computations

Forward Bearing

Bearing A to B = ?

Bearing A to B = Bearing A to F + < A

Bearing A to B = 100º00’00” + 111^{0}24’47”

Bearing A to B = 211º24’47”

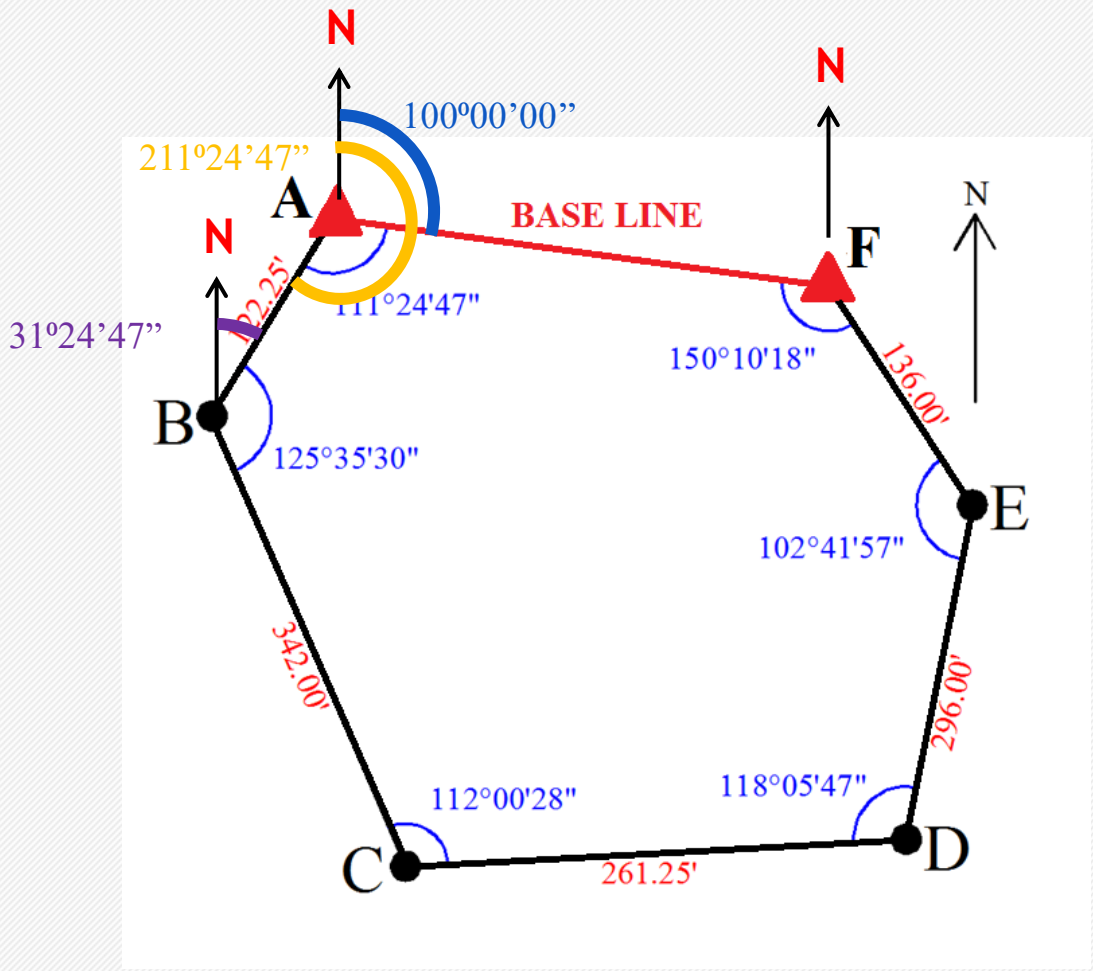
Backward Bearing

Bearing B to A = ?

Bearing B to A = Bearing A to B ± 180º

Bearing B to A = 211º24’47” - 180^{0}

Bearing B to A = 31º24’47”



Computations

Forward Bearing

Bearing B to C = ?

Bearing B to C = Bearing B to A + < B

Bearing B to C = 31º24’47” + 125^0 35’30”

Bearing B to C = 157º00’17”

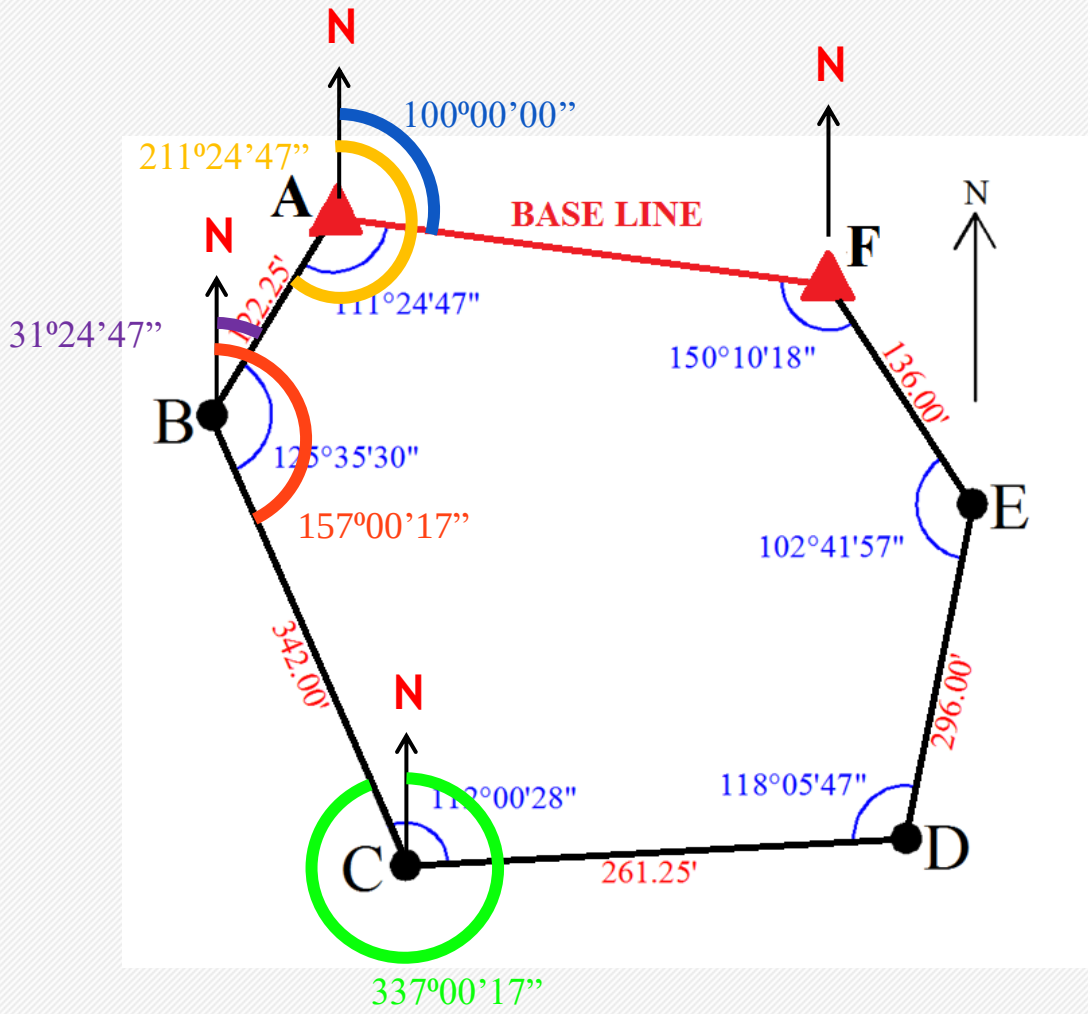
Backward Bearing

Bearing C to B = ?

Bearing C to B = Bearing B to C ± 180º

Bearing C to B = 157º00’17” + 180^0

Bearing C to B = 337º00’17”



Computations

Forward Bearing

Bearing C to D = ?

Bearing C to D = Bearing C to B + < C

Bearing C to D = 337º00’17” + 112^{0} 00’28”

Bearing C to D = 449º00’45”

Bearing C to D = 449º00’45” - 360 º

Bearing C to D = 89º00’45”

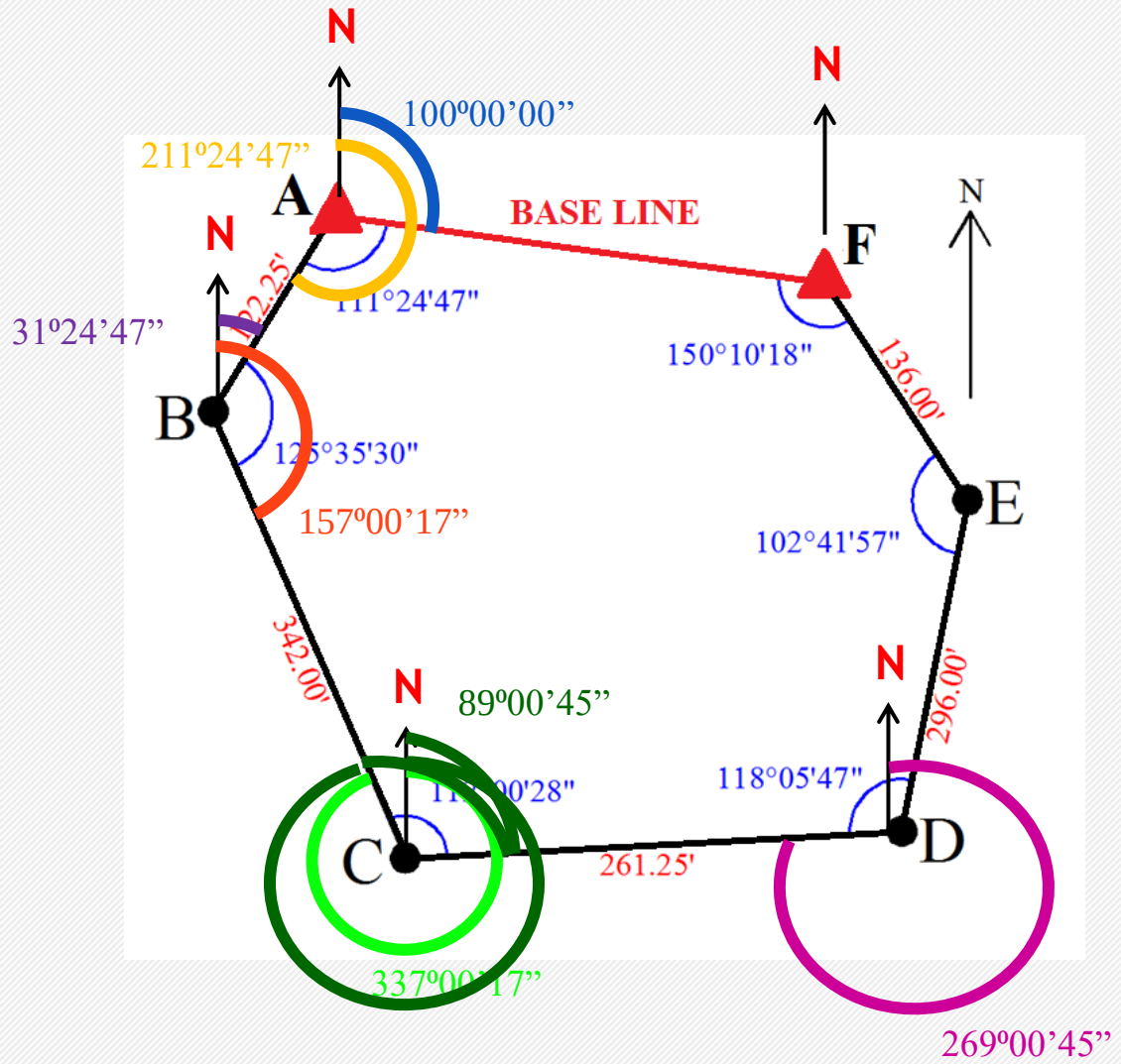
Backward Bearing

Bearing D to C = ?

Bearing D to C = Bearing B to C ± 180º

Bearing D to C = 89º00’45” + 180^{0}

Bearing D to C = 269º00’45”



Computations

Forward Bearing

Bearing D to E = ?

Bearing D to E = Bearing D to C + < D

Bearing D to E = 269º00’45” + 118^{0} 05’47”

Bearing D to E = 387º06’32”

Bearing D to E = 387º06’32” - 360 º

Bearing D to E = 27º06’32”

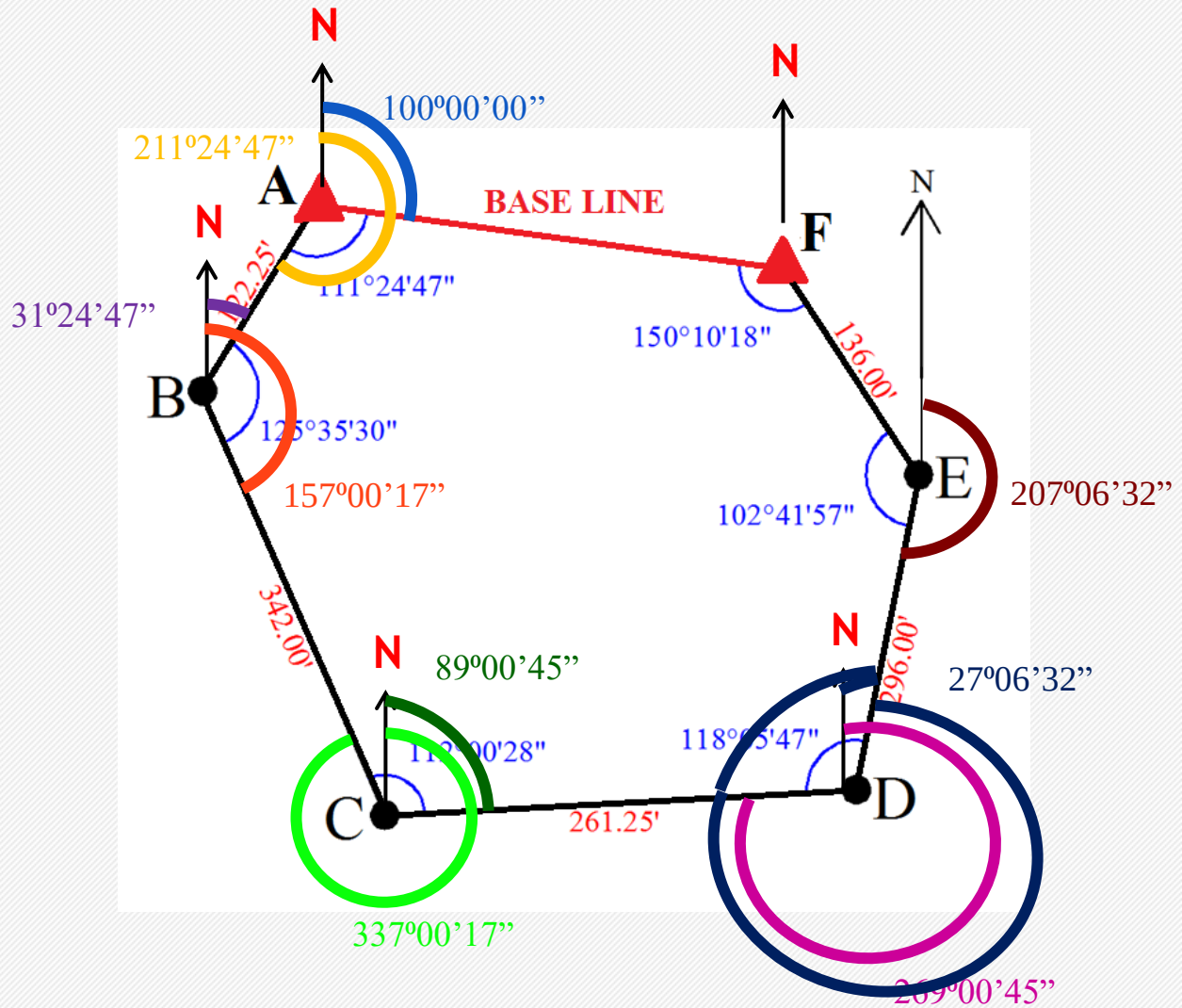
Backward Bearing

Bearing E to D = ?

Bearing E to D = Bearing D to E ± 180º

Bearing E to D = 27º06’32” + 180^{0}

Bearing E to D = 207º06’32”



Computations

Forward Bearing

Bearing E to F = ?

Bearing E to F = Bearing E to D + < E

Bearing E to F = 207º06’32” + 102^0 41’57”

Bearing E to F = 309º48’29”

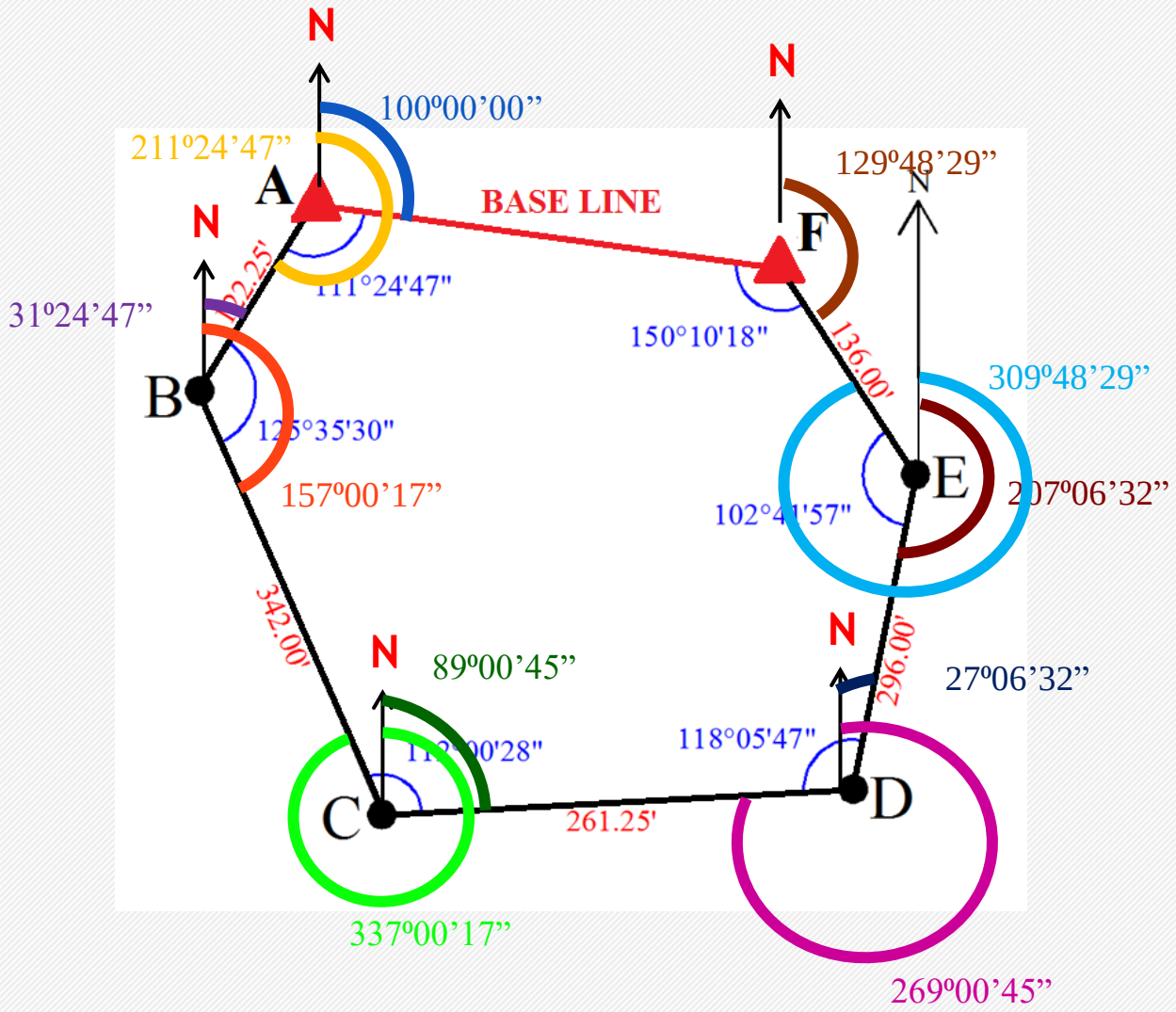
Backward Bearing

Bearing F to E = ?

Bearing F to E = Bearing E to F ± 180º

Bearing F to E = 309º48’29”- 180^0

Bearing F to E = 129º48’29”



Computations

Forward Bearing

Bearing F to A = ?

Bearing F to A = Bearing F to E + < F

Bearing F to A = 129º48’29” + 150^0 10’18”

Bearing F to A = 279º58’47”

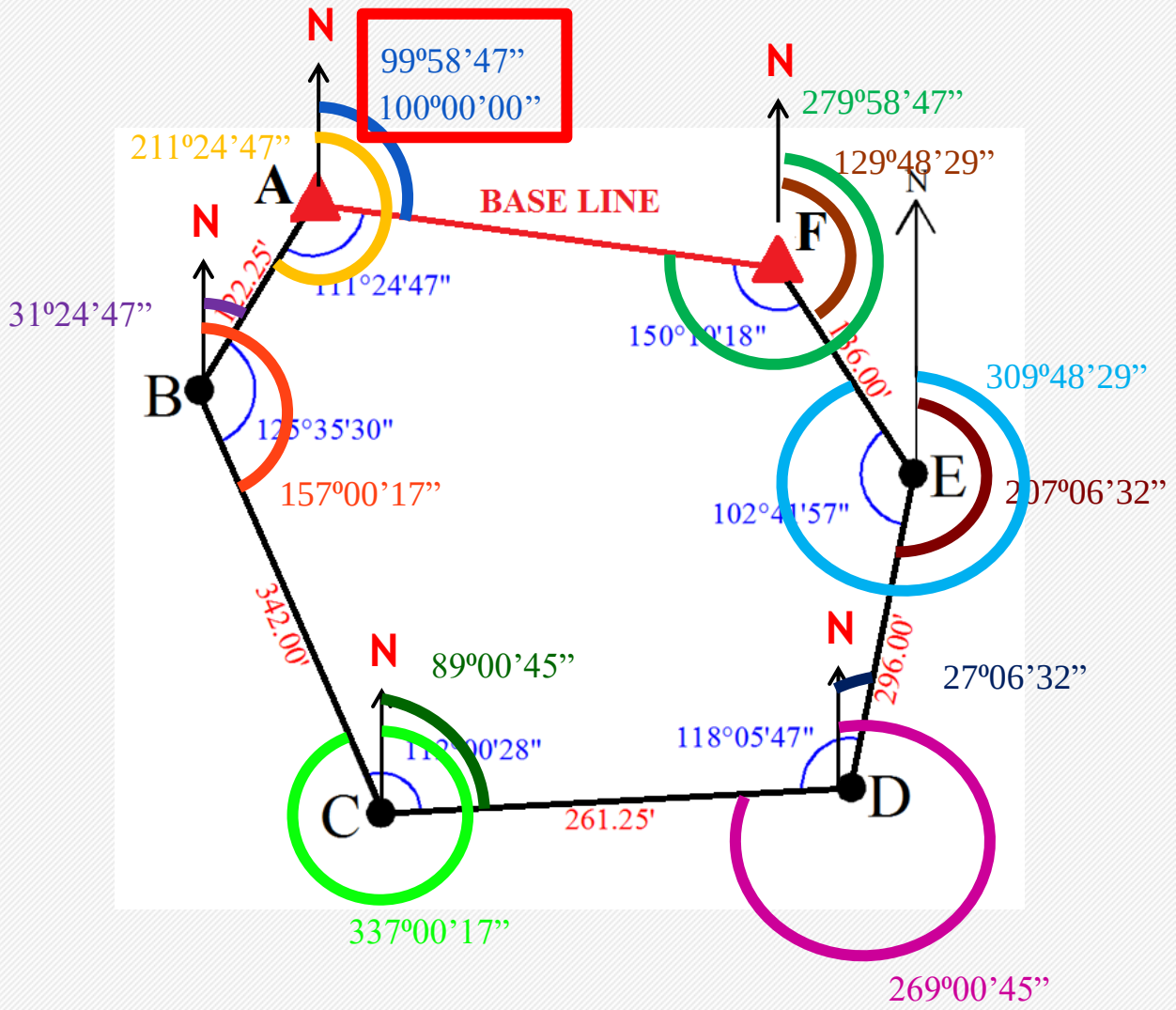
Backward Bearing

Bearing A to F = ?

Bearing A to F = Bearing F to A ± 180º

Bearing A to F = 279º58’47” -180^0

Bearing A to F = 99º58’47”



Computations

Closing Error

100º00’00”- 99º58’47” = 0º01’13”

= 73”

Note: Before error distribution, using the acceptable angular misclosure formula, check whether it is in the acceptable limits or not.

Allowable Error

$$E'' = \pm\ (K \times S \times \sqrt{n})''$$

where

K is a multiplication factor of 1 to 3 depending on weather conditions & the number of rounds taken

S is the smallest reading interval on the total station in seconds

n is the number of angles measured

Computations

$$E'' = \pm\ (K \times S \times \sqrt{n})''$$

$$E'' = \pm\ (3 \times 10'' \times \sqrt{6})''$$

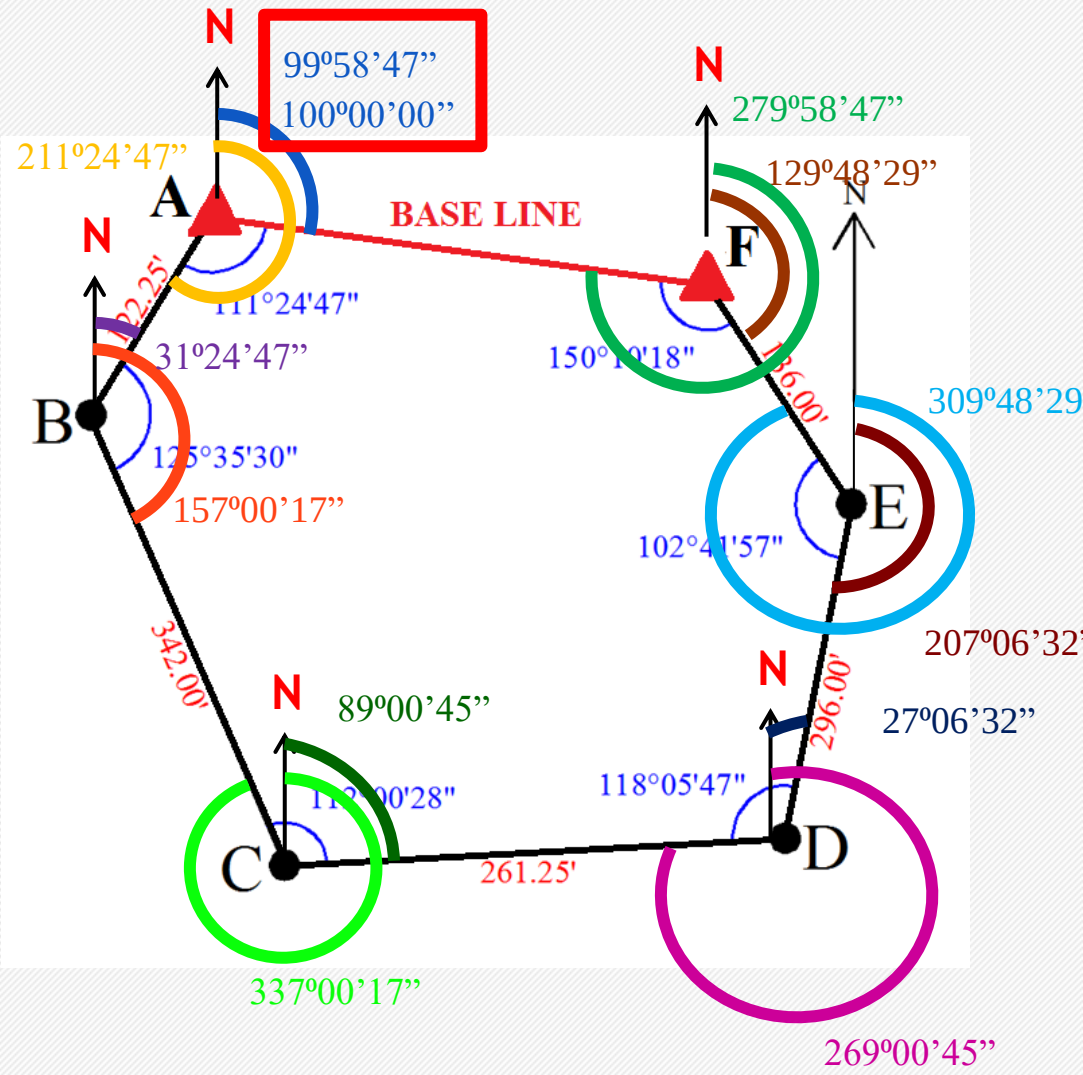
$$E'' = \pm\ 73''$$

- **If the misclosure is acceptable** (less than the allowable), divide it equally between the observed angles
- **If the misclosure is not acceptable** (greater than the allowable), the angles must be remeasured provided that no visible error/blunder can be located in the angle bookings or summation

ERROR FOR EACH
FORWARD BEARING= 12″ (excluding the last one)

Computations

Station From To	Bearing			Correction		Corrected Bearing		
	D	M	S	M	S	D	M	S
A to F	100	0	0					
A to B	211	24	47		+12	211	24	59
B to A	31	24	47					
B to C	157	00	17		+24	157	00	41
C to B	337	00	17					
C to D	89	00	45		+36	89	01	21
D to C	269	00	45					
D to E	27	06	32		+48	27	07	20
E to D	207	06	32					
E to F	309	48	29	+1		309	49	29
F to E	129	48	29					
F to A	279	58	47	+1	+13	280	0	0



Abstract

Station From To	Observed Angle			Bearing			Correction		Corrected Bearing		
	D	M	S	D	M	S	M	S	D	M	S
A to F				100	0	0			100	0	0
A to B	111	24	47	211	24	47		+12	211	24	59
B to C	125	35	30	157	00	17		+24	157	00	41
C to D	112	0	28	89	00	45		+36	89	01	21
D to E	118	5	47	27	06	32		+48	27	07	20
E to F	102	41	57	309	48	29	+1		309	49	29
F to A	150	10	18	279	58	47	+1	+13	280	0	0

Computations

Co-ordinates

$\Delta E = L \sin\theta_{A \text{ to } B}$

$\Delta N = L \cos\theta_{A \text{ to } B}$

$E_B = E_A + \Delta E$

$N_B = N_A + \Delta N$

Remember to use corrected bearings.

Co-ordinates of B

Bearing A to B = θ = 211° 24′ 59″

Distance A to B = L =122.26 ft

ΔE = (L sin θ) = -63.73 ft

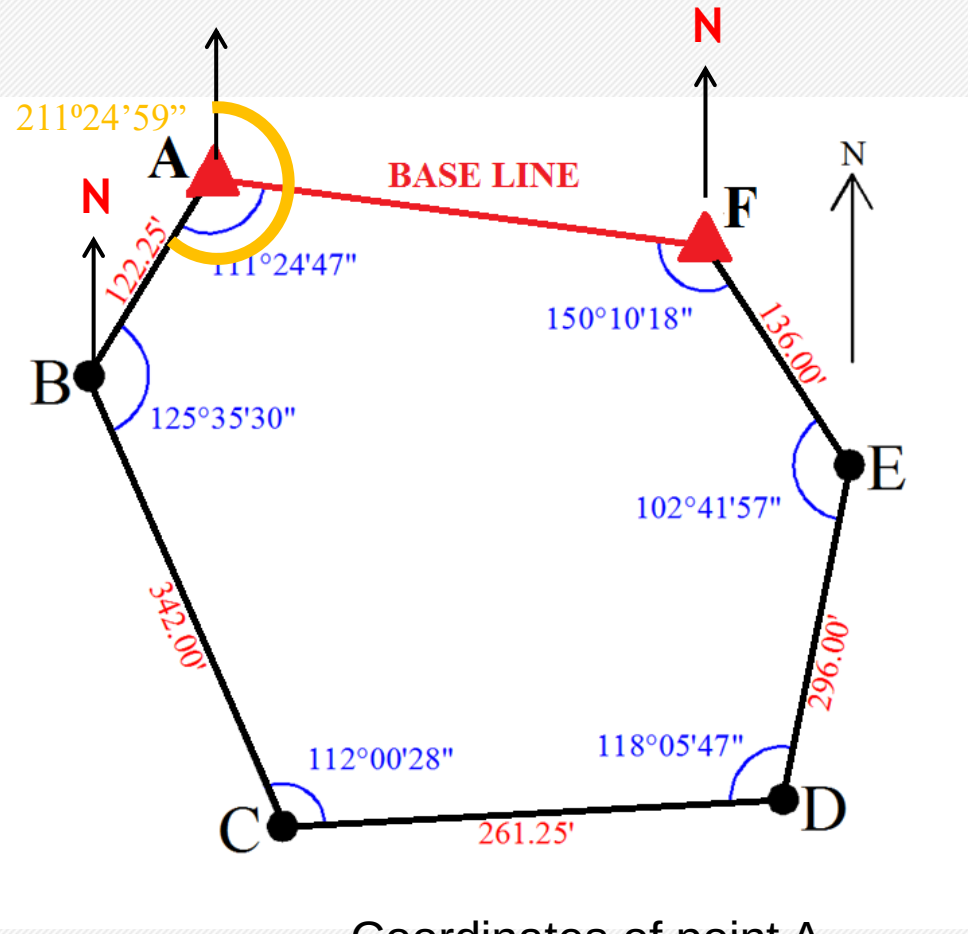
ΔN = (L cos θ) = -104.34 ft

E_B = $E_A + \Delta E$

= 4500.00 – 63.73 = **4436.27 ft**

N_B = $N_A + \Delta N$

= 5300.00 – 104.34 = **5195.66 ft**



Coordinates of point A

E_A=4500ft

N_A=5300ft

Computations

Co-ordinates of C

Bearing B to C = θ = 157° 00′ 41″

Distance B to C = L =342.00 ft

ΔE = (L sin θ) = 133.57 ft

ΔN = (L cos θ) = -314.84 ft

E_C = E_B + ΔE

= 4436.27 + 133.57 = **4569.84 ft**

N_C = N_B + ΔN

= 5195.66 – 314.84 = **4880.82 ft**

O B -63.72 FT

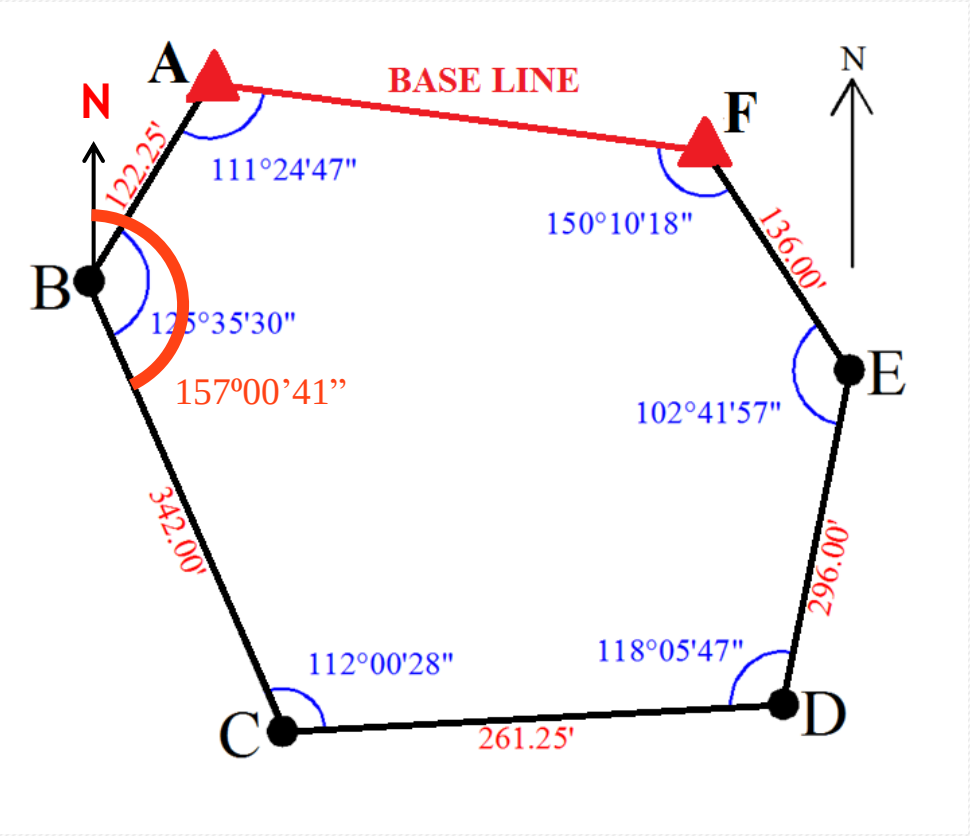
ΔN A TO B = -104.33 FT

E B = E A + ΔE B

= 4500.00 – 63.72 = **4436.28 FT**

N B = N A + ΔN B

= 5300.00 – 104.33 = **5195.67 FT**



Computations

Co-ordinates of D

Bearing C to D = θ = 89° 01′ 21″

Distance C to D = L =261.25 ft

ΔE = (L sin θ) = 261.21 ft

ΔN = (L cos θ) = 4.46 ft

E_D = E_C + ΔE

= 4569.84 + 261.21 = **4831.05 ft**

N_D = N_C + ΔN

= 4880.82 + 4.46 = **4885.28 ft**

O B -63.72 FT

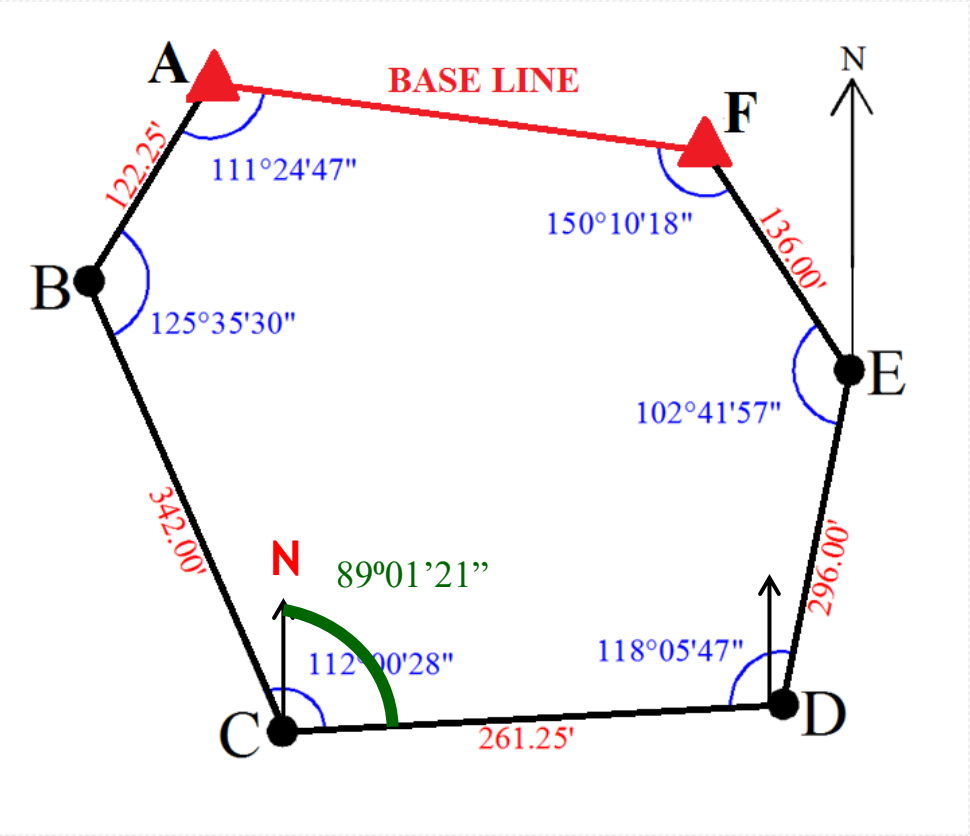
ΔN A TO B = -104.33 FT

E B = E A + ΔE B

= 4500.00 – 63.72 = **4436.28 FT**

N B = N A + ΔN B

= 5300.00 – 104.33 = **5195.67 FT**



Computations

Co-ordinates of E

Bearing D to E= θ = 27˚ 07′ 20″

Distance D to E = L =296.00 ft

ΔE = (L sin θ) = 134.94 ft

ΔN = (L cos θ) = 263.45 ft

$E_E = E_D + \Delta E$

= 4831.05 + 134.94 = **4965.99 ft**

$N_E = N_D + \Delta N$

= 4885.28 + 263.45 = **5148.73 ft**

O B -63.72 FT

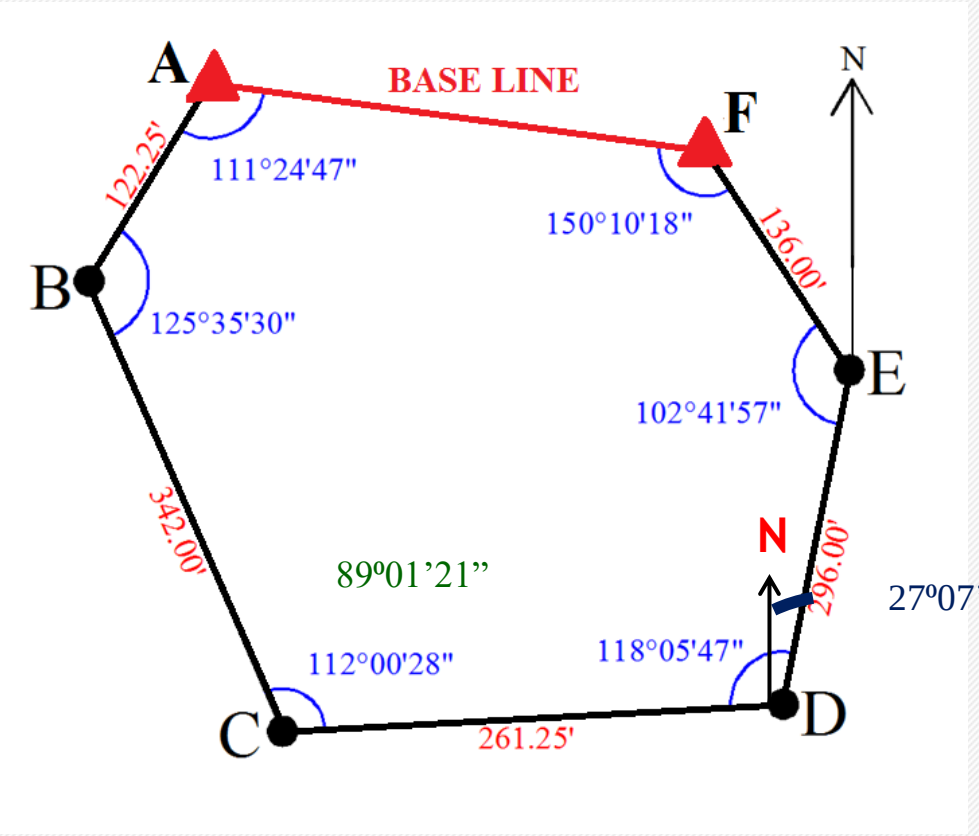
ΔN A TO B = -104.33 FT

E B = E A + ΔE B

= 4500.00 – 63.72 = **4436.28 FT**

N B = N A + ΔN B

= 5300.00 – 104.33 = **5195.67 FT**



Computations

Co-ordinates of F

Bearing E to F= θ = 309˚ 49′ 29″

Distance E to F = L =136.00 ft

ΔE = (L sin θ) = -104.45 ft

ΔN = (L cos θ) = 87.10 ft

E_F = E_E + ΔE

= 4965.99 – 104.45 = **4861.54 ft**

N_F = N_E + ΔN

= 5148.73 + 87.10 = **5235.83 ft**

O B -63.72 FT

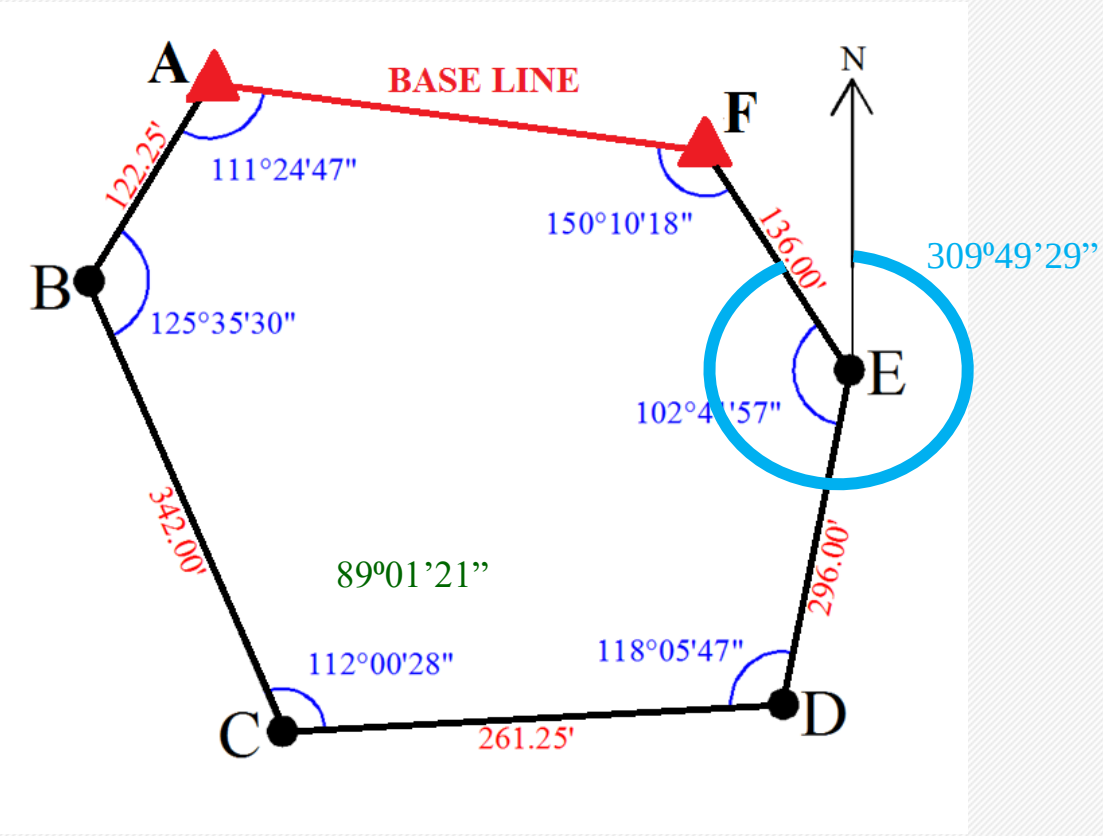
ΔN A TO B = -104.33 FT

E B = E A + ΔE B

= 4500.00 – 63.72 = **4436.28 FT**

N B = N A + ΔN B

= 5300.00 – 104.33 = **5195.67 FT**



Computations

Error Computation

Co-ordinates of F

Given	**Calculated**
E_F = 4860.63ft	E_F = 4861.54 ft
N_F = 5236.41ft	N_F = 5235.83 ft

ERROR IN EASTING

= (Known Easting of F- computed Easting of F)

=4860.63-4861.55

=-0.92 ft

Error on Per Station =0.92/5 =-0.184 ft

ERROR IN NORTHING

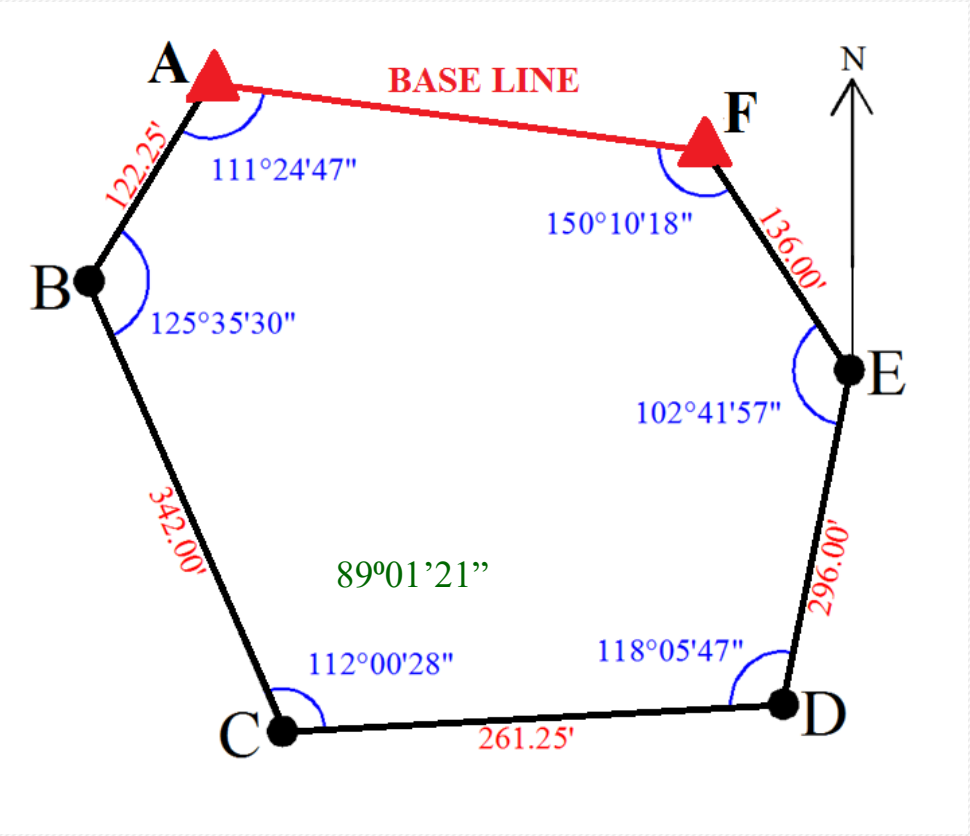
= (Known Northing of F- computed Northing of F)

=5236.41-5235.84

=0.57 ft

Error on Per Station =0.57/5 = 0.114 ft

$N_B = N_A + \Delta N_B$



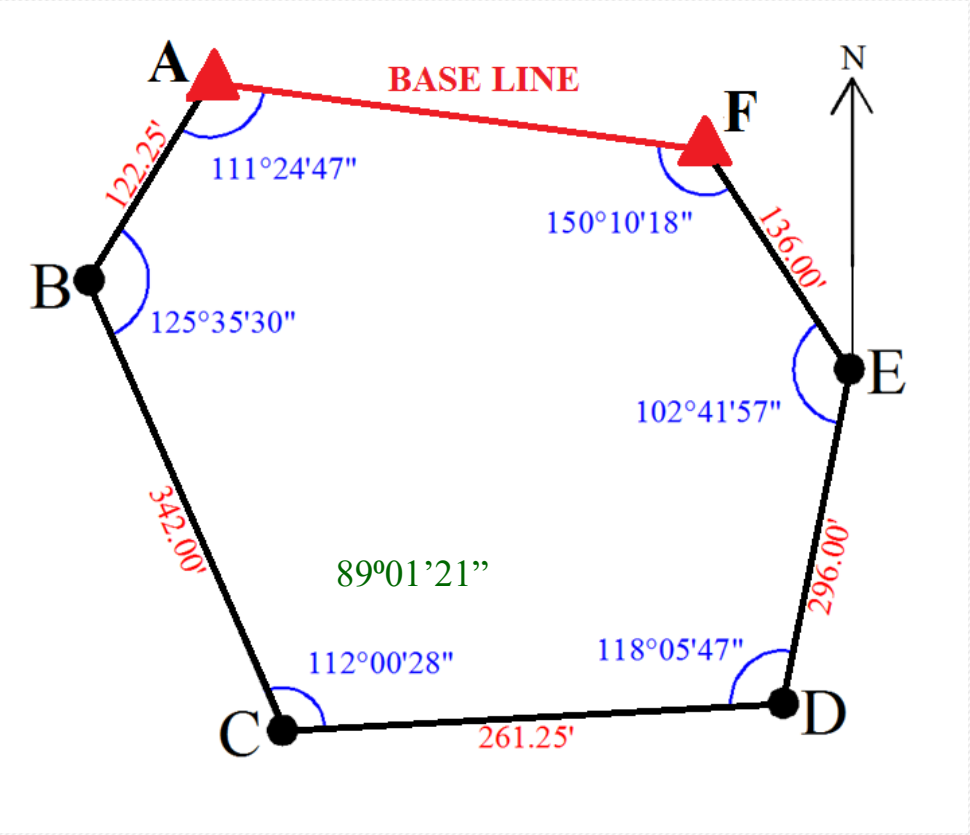
Computations

Error Adjustment

Adjustments in Eastings

Error on Per Station =0.92/5 =-0.184 ft

Station No.	Correction multiplication	Correction After round off
B	-0.184 x 1 = 0.184	-0.18
C	-0.184 x 2 = 0.368	-0.37
D	-0.184 x 3 = 0.552	-0.55
E	-0.184 x 4 = 0.736	-0.74
F	-0.184 x 5 = 0.92	-0.92



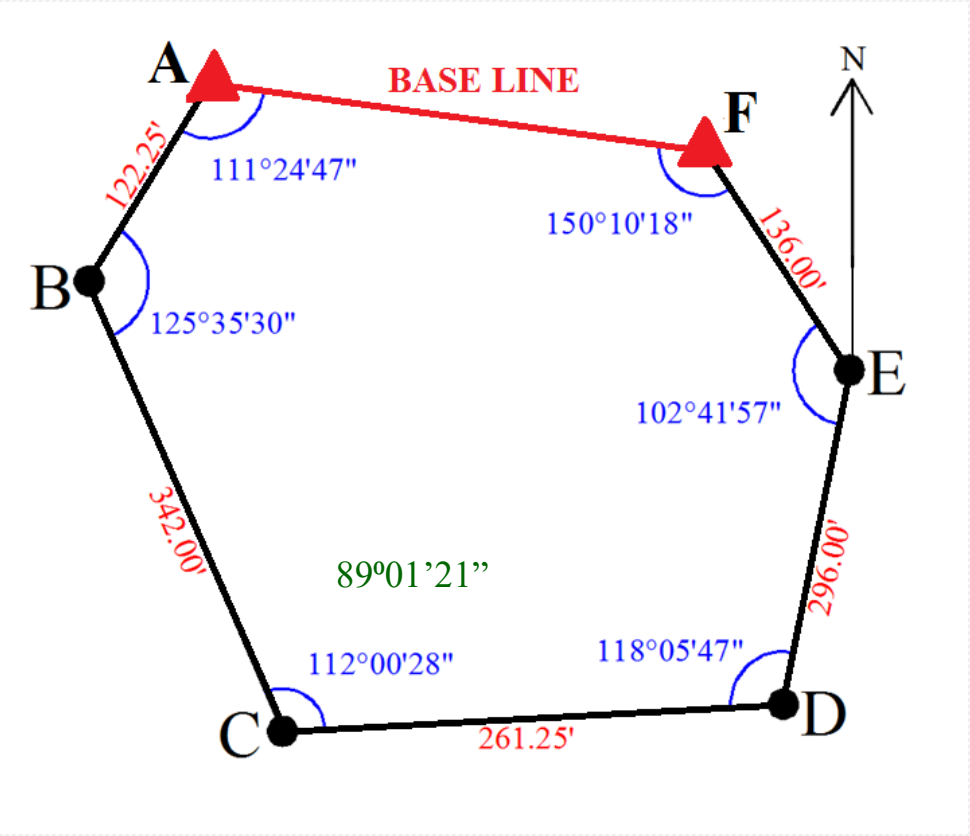
Computations

Error Adjustment

Adjustments in Northings

Error on Per Station =0.57/5 = 0.114 ft

Station No.	Correction multiplication	Correction After round off
B	0.114x 1 = 0.114	0.11
C	0.114 x 2 = 0.228	0.23
D	0.114 x 3 = 0.342	0.34
E	0.114 x 4 = 0.456	0.46
F	0.114 x 5 = 0.57	0.57



Computations

Fractional Linear Misclosure

***F.L.M* = 1 in ∑D/e**

where

e is the linear misclosure [= $\sqrt{(e_E^2 + e_N^2)}$]

$\sum D$ is the total length of the traverse

Note: *The FLM is always computed for a traverse before adjustments to the coordinate differences and the station coordinates are calculated – this is compared to traverse specifications (shown on next slide) and a decision is made to accept or reject the traverse.*

Computations

Here:

e i.e. linear misclosure = $\sqrt{(e_E^2 + e_N^2)} = \sqrt{[(-0.92)^2 + (0.57)^2]} = 1.0822$ ft

$\sum D$ i.e. the total length of the traverse = 1523.5 ft

F.L.M **= 1 in ∑D/e**

F.L.M **= 1 in 1523.5/1.0822**

= 1 in 1400 (approx.) …… *Low accuracy traverse; acceptable from point of view of training of students*

Computations

Type	Typical Accuracy	Purpose	Angular Measurement	Distance Measurement
Geodetic or precise	1 in 50,000 or better	Provision of very accurate reference points for engineering surveys/Primary control points	Use 0.1” industrial theodolite	Accuracy better than 1 mm required – use specialised electronic distance measuring equipment or industrial total station
General	1 in 5000 t0 1 in 50,000	General engineering work including site surveys & setting out of structures & roadworks/Adding secondary control to primary control networks	Electronic theodolite or total station with accuracy between 1” and 10” required	Use total station or steel tapes to match accuracy requirement
Low accuracy	1 in 500 to 1 in 5000	Control for drainage schemes and earthworks/Small-scale mapping	Electronic or optical theodolite of 20” or lower accuracy required	Total station, steel or synthetic tapes can be used

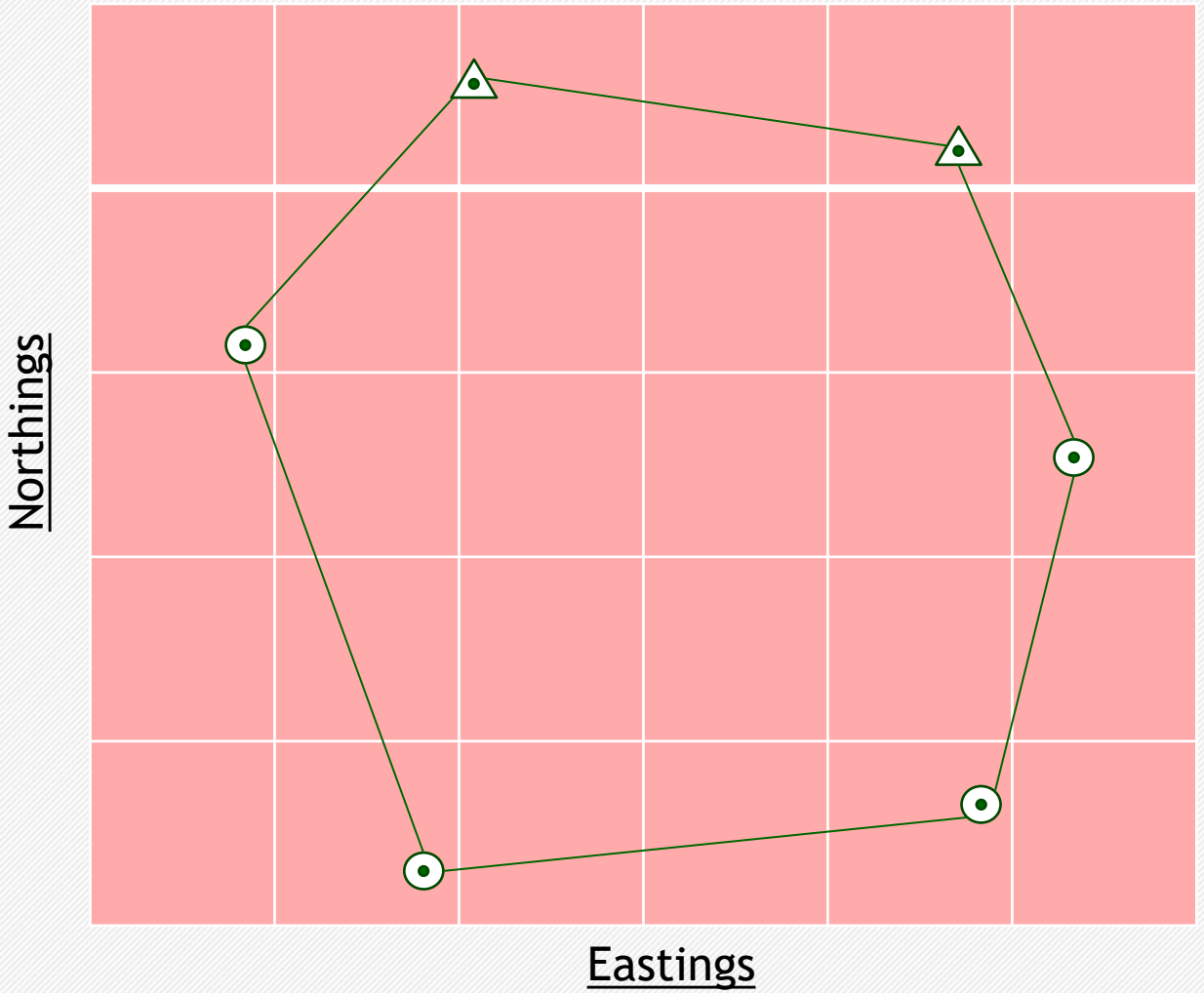
Computations

Error Adjustment

Station	Uncorrected Co-ordinates		Corrections		Corrected Co-ordinates	
	Eastings	Northings	e	n	Eastings	Northings
A	4500	5300			4500.00	5300.00
B	4436.27	5195.66	-0.18	0.11	4436.09	5195.77
C	4569.84	4880.82	-0.37	0.23	4569.47	4881.05
D	4831.05	4885.28	-0.55	0.34	4830.50	4885.62
E	4965.99	5148.73	-0.74	0.46	4965.25	5149.19
F	4861.54	5235.83	-0.92	0.57	4860.62	5236.40

Computations

Plotting



Scale
1 inch = 30 feet

Essentials for Next Week

Submissions

1. Traverse Computations
2. Report
3. Traverse Points plotted on a graph sheet(1 inch = 10 divisions) on a scale 1 inch = 30 feet

Bring the following:

1. 1 x Scholar's Sheet
2. T-Square
3. Set Squares
4. Pencil (2H), erasers, sharpeners
5. Paper Tape
6. Black Pointer (0.3 mm)
7. A-4 pages